SMOOTHING CALABI–YAU THREEFOLDS WITH NODES AND DEL PEZZO CONE POINTS

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ABSTRACT. A Calabi–Yau threefold whose singularities are ordinary double points and anticanonical cones over del Pezzo surfaces of degree 6 or 7 has smoothable germs but need not be smoothable, and Friedman's nodal criterion does not apply, because a del Pezzo cone point has no small resolution. For hypersurfaces in Gorenstein toric Fano fourfolds we give a combinatorial condition under which the ambient toric mechanism cannot smooth such a threefold, and settle every polytope with at most nine vertices.

The obstruction combines a local grading theorem with a combinatorial extension test. Altmann's calculation reduces all one-sided degrees that can move a degree-7 cone to its Gorenstein-degree affine line; a condition on a facet, called locking, depends only on the face lattice and blocks that line. The remaining mixed-sign degrees retain a singular curve, which every anticanonical Cartier divisor must meet. Against this obstruction we place a nine-vertex reflexive polytope whose hypersurface has two nodes, neither of which lies in any relation among the nodal exceptional curve classes, and one degree-7 cone point, and which we prove smoothable, so that no purely nodal mechanism produces that smoothing. Between them these settle that range. Of the reflexive 4-polytopes with at most nine vertices whose hypersurface has only nodes and degree-6 or degree-7 cone points, all isolated, exactly 77 carry a pentagonal 2-face; running the companion's criterion over them leaves 76 non-smoothable, and the exception is the polytope above.

We also record a first order converse to the companion paper's obstruction, proved by reduction to Friedman–Laza: a relation among the link classes that at every germ is nonzero and dies in the Milnor fibre produces a first order smoothing. The two halves of the criterion then meet, and what separates them is integrability alone, an unobstructedness statement for isolated canonical Gorenstein singularities that are not complete intersections.

1. Introduction

1.1. **The question.** A projective Calabi–Yau threefold with isolated singularities, each of which is smoothable as a germ, need not be smoothable. For ordinary double points the precise mechanism is classical. Friedman [Fri86] showed that a nodal Calabi–Yau threefold X admitting a small resolution $\pi\colon \widehat{X} \to X$ (from Section 2 onward $\widehat{X}$ will denote a crepant resolution, which at a node is small and at a cone point is not) is smoothable as soon as the classes of the exceptional curves $C_1, \dots, C_k \subset \widehat{X}$ satisfy a relation

$$\sum_{j=1}^{k} \lambda_j\,[C_j] = 0 \quad \text{in } H_2(\widehat{X};\mathbb{Q}), \qquad \lambda_j \neq 0 \text{ for all } j. \tag{1}$$

Date: 24 August 2026.

2020 *Mathematics Subject Classification.* 14J32 (primary); 14B07, 14M25, 14J45, 52B20 (secondary).

Key words and phrases. Calabi–Yau threefolds, smoothing of singularities, del Pezzo cones, ordinary double points, Minkowski decomposition, T-varieties, reflexive polytopes.

The converse is not in [Fri86]; it is due to Namikawa [Nam02, Thm. 2.5]. Tian, Kawamata and Ran gave other proofs of the sufficiency, and Namikawa extensions; see [Nam94, NS95, RT09] and, for the modern deformation theory of the singular varieties involved, [FL25, FL24].

Once the singularities are not all nodes, both halves become harder, and for different reasons. The germs that arise after the node in the toric setting are the *anticanonical cones over del Pezzo surfaces*, $\mathrm{Cone}(\mathrm{dP}_d)$, obtained by coning the anticanonical embedding; the relevant degrees here are $d = 7$ and $d = 6$. Such a germ is Gorenstein, canonical, isolated and smoothable, and its versal deformation was computed by Altmann [Alt95, Alt97]: a line with an embedded point for $d = 7$, a line meeting a plane for $d = 6$. What it does not have is a small resolution: the crepant resolution is divisorial, with a del Pezzo *surface* over the point. So there is no exceptional curve whose class could enter a relation like (1), and Friedman's argument has no place to start.

In the companion paper [Wue26b] the necessity half was proved for a *mixed* inventory of nodes together with dP_6 and dP_7 cone points, by replacing the missing curve class at a dP_7 point with a canonical line inside $K^\perp$ in the del Pezzo lattice. The resulting criterion is a finite linear-algebra test, and it obstructs: it produces Calabi–Yau threefolds, with every germ locally smoothable, that admit no smoothing at all. But a necessity criterion is silent whenever its test passes, and it passes often. The present paper asks what happens then.

The same case is left open from the other side. For a reflexive Δ with unit edges, the germs of X over a non-smooth 2-face F are $\ell(F^\circ)$ copies of the cone over F, where $\ell(F^\circ)$ is the lattice length of the dual edge. When $\ell(F^\circ) \geq 2$ those germs can be smoothed a face at a time, by deformations of the ambient toric pair [Wue26a]. When $\ell(F^\circ) = 1$, which is the second condition of Definition 2.4 and the setting of this paper, that mechanism has nothing to act on, and [Wue26a, Prop. C] shows why no general theorem applies either: the crepant contraction of the exceptional del Pezzo is never primitive and X is not $\mathbb{Q}$-factorial at the point, so neither Friedman's criterion nor Gross's theorems decide the case. It is posed there as an open question, and it is the question this paper answers.

1.2. **Results.** Our threefolds are Batyrev hypersurfaces [Bat94]: $\Delta \subset N_{\mathbb{R}} \cong \mathbb{R}^4$ a reflexive polytope, Σ its face fan, $\mathbb{P}_\Delta = \mathrm{TV}(\Sigma)$ the Gorenstein toric Fano fourfold, and $X \in |-K_{\mathbb{P}_\Delta}|$ generic. The singularities of X are read off the 2-faces of Δ (Section 2).

There is exactly one deformation mechanism available uniformly across this class, and it is toric. A Minkowski decomposition of a suitable cross-section of Σ deforms $\mathbb{P}_\Delta$ itself; if the hypersurface can be made to follow, the germs of X are deformed along with the ambient germs. For affine toric varieties this is Altmann's construction [Alt95, Alt97]; for the non-affine, non-simplicial ambients we need, it is the theory of Ilten–Vollmert [IV12], who realise the deformation as a family of complexity-one T-varieties. We call the resulting object a *single-degree family*; Definition 3.3 makes it precise.

Theorems A and B say that this mechanism genuinely cannot and genuinely can, and say so for reasons one reads off the polytope; Theorem C settles the pentagonal part of the small range outright; and Theorem D, which is not about the toric mechanism at all, supplies the missing half of the criterion the first three are measured against.

Informally: a facet carries an *edge-forcing propagation*, a closure rule on its edge graph that reads only the face lattice and which of its 2-faces are triangles, and Definition 4.12 makes it precise. A set of edges tied together by the propagation is called *locked*, and a facet is locked when its whole edge set is. For a pentagonal face G, Altmann's grading theorem first reduces the infinite degree lattice to the affine line on which every ray of G has height one. When the two containing facets are locked and their positive ranges cover this line, no admissible Minkowski decomposition can move G.

Theorem A (= Theorems 5.1 and 5.2)**.** *Let Δ be a reflexive 4-polytope in the admissible framework and let G be a unit-edge pentagonal 2-face, contained in facets F_1, F_2. Let*

$$\mathcal{A}_G = \{r \in M : \langle r, v\rangle = 1 \text{ for every } v \in V(G)\}.$$

Suppose both facets are locked and every $r \in \mathcal{A}_G$ is positive on every vertex of F_1 or on every vertex of F_2. Then for every primitive degree $R \in M$ the general ambient fibre of every single-degree family at R has a singular curve. Consequently no relative anticanonical effective Cartier family inside such a single-degree ambient family has smooth general fibre. The hypothesis is decidable.

The criterion applies to the reflexive polytope Δ_{19} with 19 vertices recorded in (7), whose generic anticanonical hypersurface X_{19} has fourteen nodes and one dP_7 cone point, all isolated. So no single-degree family smooths X_{19} through the relative anticanonical system.

The node relation system of X_{19} has two coloops [Wue26b, §9.5], so any first-order smoothing must recruit the dP_7 direction. Theorem A proves more for the toric mechanism: every single-degree ambient fibre retains a singular curve that traps its anticanonical divisors.

Theorem B (= Theorem 7.12)**.** *Let*

$$\Delta_9 = \mathrm{conv}\big\{e_1,\ e_2,\ -e_1-e_2,\ e_3,\ e_4,\ (-4,-2,-1,0),\\ (-4,-2,0,-1),\ (3,1,1,1),\ (2,1,1,1)\big\} \subset \mathbb{Z}^4, \tag{2}$$

a reflexive polytope with nine vertices and twelve facets, and let X_9 be its generic anticanonical hypersurface. Then

(1) $\mathrm{Sing}(X_9)$ *consists of exactly three points: two ordinary double points and one* dP_7 *cone point;*
(2) *both nodes are coloops of the node relation system, so no purely nodal mechanism smooths X_9, and the criterion of* [Wue26b] *is silent on it;*
(3) X_9 is *smoothable. Explicitly, the single-degree family at $R = (0, 1, -1, -1)$, for the Minkowski decomposition of Table 2, has smooth general anticanonical member.*

Part (iii) is the point. The two nodes being coloops means that no relation (1) exists among the exceptional curves of a small resolution of the nodes alone; whatever smooths X_9 must move the dP_7 point at the same time. One family does, and one can watch it do so: at the two nodes the cross-section splits as primitive segment + primitive segment, at the dP_7 point as primitive segment + unimodular triangle, which is Altmann's unique nontrivial lattice decomposition of the pentagon [Alt95, (7.1.4)].

Restricting to the admissible framework makes the range small enough to settle outright.

Theorem C (= Theorem 6.3)**.** *There are exactly 77 reflexive 4-polytopes with at most nine vertices that lie in the admissible framework of Definition 2.4, whose 2-faces are unimodular triangles, unit squares and unit-edge pentagons or hexagons with one interior point and dual edge of length one, and carry a pentagonal 2-face. For 76 of them the criterion of* [Wue26b] *obstructs, so X admits no one-parameter smoothing and hence, by* [Wue26b, Cor. 6.7]*, is not smoothable over any base germ; 67 of these have a single singular point. The remaining one is Δ_9, and X_9 is smoothable. Exactly 76 of the 77 pentagons are locked, the unlocked one being Δ_9's.*

The criterion of [Wue26b], Theorem A and the construction therefore decide every polytope in the range, and the last two agree: in it the ambient construction is available precisely where the criterion permits a smoothing.

Finally, the criterion of [Wue26b] is a *necessary* condition, and the statement it is one half of is a mixed analogue of Friedman's. We prove the other half at first order.

Theorem D (= Theorem 8.8)**.** *Let X be in the admissible framework of Definition 2.4, so that every 2-face of its polytope is a unimodular triangle, a unit square, or a unit-edge pentagon or hexagon with one interior point and dual edge of lattice length one, and X therefore has only nodes and degree-7 or degree-6 cone points, all isolated. Suppose $\omega_X \cong \mathcal{O}_X$, $h^1(\mathcal{O}_X) = 0$, and the $\partial\bar{\partial}$-lemma for a resolution. Suppose there is a class*

$$\kappa \in \ker\Big(\bigoplus_p H_2(L_p;\mathbb{Q}) \longrightarrow H_2(\widehat{X};\mathbb{Q})\Big)$$

whose component at every singular point is nonzero and lies in $\mathcal{R}_p$, the part of the link homology that dies in the Milnor fibre. Then X admits a first order smoothing.

Theorem D is proved by a local-to-global support sequence. The local Hodge comparison is established separately for nodes and del Pezzo cones, and an equivariant calculation identifies the smoothing branches with the prescribed link-homology subspaces.

That hypothesis is exactly the criterion of [Wue26b], so the criterion and its converse meet at first order and the gap between them is integrability alone.

We name that gap precisely, because it is not a deficiency of the present argument. Passing from a first order smoothing to an actual one is, in every treatment we know, the hypothesis that $\mathrm{Def}(X)$ be unobstructed, and the unobstructedness inputs of the Friedman–Laza route require the singularities to be local complete intersections, and a cone over a del Pezzo surface of degree $d > 4$ is never one [Gro97, (3.3)]. Routes that do reach non-lci germs exist, [Gro97, Thm. 5.8] and [Nam97, Cor. 6] among them, but each requires $\mathbb{Q}$-factoriality or the primitivity that supplies it, and §8 shows the admissible class has neither. Supplying an unobstructedness statement for isolated canonical Gorenstein singularities that are not complete intersections would close the first route; it lies beyond the open problems of [FL25, Summary and Outlook], which stop at the local complete intersection case. The asymmetry is structural: the necessity half of [Wue26b] escapes the complete-intersection barrier by being topological, a period computation that never touches deformation theory, and a sufficiency argument has no such escape because it must produce a family.

One case is settled and one is not. When X is $\mathbb{Q}$-factorial the localisation map is surjective and Namikawa's theorem covers the whole inventory (Theorem 8.2); by [Wue26b, Prop. 8.4] that is exactly where the relation kernel is everything. The block in question is the set of rows of [Wue26b]'s matrix contributed by one germ, one row per $K^\perp$ functional, which by Section 8.1 is a basis of the germ's T^1; we call it that germ's *channel block*. It has maximal rank at every polytope of Theorem C and at Δ_{19} as well, so no singular X in the admissible framework carrying one of these germs is $\mathbb{Q}$-factorial and that case, although settled, is empty here. The open content of the conjecture is the singular part of the framework itself. Applied to Δ_{19}, whose criterion is silent, Theorem D says X_{19} has a first order smoothing that Theorem A shows no single-degree family of the ambient can produce.

1.3. **Why the criterion has to be combinatorial.** The mechanism is indexed by a degree $R \in M = \mathrm{Hom}(N,\mathbb{Z})$, and M is infinite. A statement quantified over all of it needs a structural reduction; without one, checking the familiar Gorenstein-degree line says nothing about the rest.

The direct repair is to compute, for each degree, the dimension of the summand cone of the relevant cell of the slice complex, and to check it is one. That does not work. *The dimension of the summand cone is not an invariant of the cone* σ_F: on Δ_{19} the facet numbered 12 has Minkowski summand cone of dimension 2 at the lattice facet and dimension 1 at other cross-sections of the same σ_{12} (Remark 4.8). A criterion phrased in that language is degree-dependent by nature.

The first reduction is local and deformation-theoretic. Altmann's calculation of T^1 shows that every one-sided degree off the Gorenstein-degree line gives a trivial admissible deformation of the dP_7 chart (Lemma 4.6); complete-locus degrees retain a singular curve by Lemma 4.1. What remains on the line is degree-independent and combinatorial. A cross-section of a pointed cone by a transversal hyperplane is a polytope combinatorially isomorphic to the base (Lemma 4.9); the edge directions move, but the three forcing rules of Proposition 4.10 use only that the edge vectors of a 2-face sum to zero and that two consecutive edges of a convex polygon are independent. Both survive rescaling. A facet on which those rules propagate to all edges is called *locked*, and a locked facet has rigid cell at every degree at once. Theorem 5.1 is the resulting criterion.

The sign lemma (Lemma 4.15) says that along the resulting affine line the two facets containing the pentagon have complementary monotone ranges of positivity. Their thresholds are finite data. On Δ_{19} the check is especially sharp: the line is $(-1, s, -1-s, 0)$, and the two ranges are $s \geq 0$ and $s \leq -1$.

1.4. **How special the two phenomena are.** Neither theorem would mean much if its polytope were a curiosity. Section 6 runs the combinatorial core of both over the Kreuzer–Skarke classification of reflexive 4-polytopes [KS00]. Two findings.

The locking half of the criterion is common but not automatic. Of the 12 508 pentagonal 2-faces occurring in reflexive 4-polytopes with at most nine vertices, 9 466 are locked, a little over three quarters, and 3 005 sit in a facet that decomposes, where Theorem 5.1 says nothing. The census does not test the separate line-coverage condition (C).

The combination Theorem B needs is, by contrast, rare. Among nine-vertex polytopes there are 11 596 pentagonal 2-faces; 63 lie in a polytope whose whole germ inventory lies in the framework, 2 886 lie in a facet that decomposes, and *exactly one* does both. That one is Δ_9.

Appendix A records two further routes that turn out to have no input on these polytopes, though both look promising from a distance. Mavlyutov's non-polynomial deformations of Batyrev hypersurfaces [Mav04, Mav05] are counted by $\sum \ell^*(\Gamma)\,\ell^*(\Gamma^*)$ over codimension-two faces, with Γ^* an edge of Δ; on a polytope all of whose edges have lattice length one, which is exactly the isolated-singularity condition, every term vanishes. And the global Minkowski decompositions his §7 takes as input do not exist here: Δ_{19} is Minkowski-indecomposable (Proposition A.2).

1.5. **What is not proved.** Theorem A does *not* say that X_{19} is non-smoothable. It says that one mechanism, at one degree at a time, does not reach it. A multi-degree family, or a deformation not induced from the ambient at all, remains possible, and the criterion of [Wue26b] is silent on X_{19}, so the expectation is that X_{19} *is* smoothable and that Theorem A is telling us the ambient is the wrong place to look. Section 8 explains why this makes X_{19} the sharpest available test of the mixed criterion.

Theorem D gives a first order smoothing and nothing more. We do not prove that it integrates, and integration is not a formality here, although the example usually cited against it does not reach our germs: Gross's non-reduced deformation space [Gro97r, Ex. 2.19] is built from cones over $\mathbb{F}_1$, which do not smooth at all. The remaining gap is the unobstructedness statement isolated in Remark 8.9.

1.6. **Verification.** Every numerical assertion below is produced by a script in the accompanying repository, listed in Appendix C with the statement it certifies. Each script computes its input from the polytope's vertices, several are implemented twice in different systems, and the two that use published formulas are validated against the worked examples of their sources. Where a step rests on a statement whose published proof we could not locate, Appendix B says so explicitly.

Acknowledgements. The question addressed here was posed to the author by his thesis advisor.

2. Germs from reflexive polytopes

We fix notation and recall the dictionary between 2-faces of a reflexive polytope and the singularities of its Calabi–Yau hypersurface. Everything in this section is standard; [Bat94] and [CLS11] are the references.

We also fix the objects the companion supplies, so that this paper can be read without it. For a singular point p of X write L_p for the *link* of the germ, $\mathcal{M}_p$ for the Milnor fibre of the induced local smoothing (so that $\mathcal{R}_p$ below depends on the smoothing, not on the germ alone), and $\widehat{X} \to X$ for a crepant resolution, which is small over a node and contracts an exceptional del Pezzo surface to each cone point. Set $\mathcal{R}_p = \ker\big(H_2(L_p;\mathbb{Q}) \to H_2(\mathcal{M}_p;\mathbb{Q})\big)$. At a node this is all of $H_2(L_p;\mathbb{Q}) \cong \mathbb{Q}$. At a cone point p the exceptional surface $E_p \subset \widehat{X}$ is a del Pezzo of some degree d, the link is the unit circle bundle of $\mathcal{O}_{E_p}(K_{E_p})$, and its bundle projection identifies $H_2(L_p;\mathbb{Q})$ with $K^\perp \subseteq H_2(E_p;\mathbb{Q})$, of rank $9-d$; in those coordinates $\mathcal{R}_p$ is the unique (-2)-line of $K^\perp$ when $d=7$, and the A_1 line or the A_2 plane of $K^\perp \cong A_1 \oplus A_2$ when $d=6$, according to the component of the miniversal base through which the germ is smoothed [Wue26b, Prop. 2.1, Lem. 4.2].

Theorem 2.1 (mixed necessity; [Wue26b, Thm. 6.1])**.** *Let X be a compact complex threefold with $\omega_X \cong \mathcal{O}_X$ and $h^1(\mathcal{O}_X)=0$ whose singular points are ordinary double points and anticanonical cones over dP_6 or dP_7, and suppose X admits a one-parameter smoothing. Set*

$$K(X) = \Big\{\kappa \in \ker\Big(\bigoplus_p H_2(L_p;\mathbb{Q}) \to H_2(\widehat{X};\mathbb{Q})\Big) \;:\; \kappa_p \in \mathcal{R}_p \text{ at every cone point } p\Big\}.$$

Then for every node q there is $\kappa \in K(X)$ with $\kappa_q \neq 0$; for every dP_7 point q there is $\kappa \in K(X)$ whose q-component is a nonzero element of the (-2)-line $\mathcal{R}_q$; and for every dP_6 point q smoothed through the one-parameter component there is $\kappa \in K(X)$ whose q-component is a nonzero element of the A_1 summand $\mathcal{R}_q$.

The step from "no one-parameter smoothing" to "not smoothable" is also the companion's, and we quote it because Theorem 6.3 rests on it.

Theorem 2.2 ([Wue26b, Cor. 6.7])**.** *Let X be as in Theorem 2.1 and admit no one-parameter smoothing. Then no fibre of a small representative of the miniversal deformation of X is smooth; consequently X is not smoothable over any base germ, and neither is any small deformation of X. Likewise no formal arc in the miniversal deformation meets all the local smoothing loci.*

The mechanism is that a smoothing over an arbitrary base germ would, by the curve selection lemma applied to the discriminant in the miniversal base, produce a one-parameter one; ruling out the one-parameter case therefore rules out all of them.

We say the criterion *obstructs* X when some coordinate at which Theorem 2.1 asserts nonvanishing vanishes identically on $K(X)$, so that no such κ exists and X has no one-parameter smoothing; and that it is *silent* when none does. Theorem 8.8 is the converse of Theorem 2.1 at first order.

Convention 2.3. $N \cong \mathbb{Z}^4$ is a lattice, $M = \mathrm{Hom}(N, \mathbb{Z})$ its dual, $\langle \cdot, \cdot \rangle$ the pairing. $\Delta \subset N_{\mathbb{R}}$ is a *reflexive* polytope: a lattice polytope with the origin in its interior whose polar $\Delta^\circ = \{u \in M_{\mathbb{R}} : \langle u, v\rangle \geq -1 \ \forall v \in \Delta\}$ is again a lattice polytope. Σ is the face fan of Δ, whose cones are $\sigma_F = \mathrm{Cone}(F)$ for F a face, and $\mathbb{P}_\Delta = \mathrm{TV}(\Sigma)$ the associated toric fourfold, which is Gorenstein Fano. For a facet F we write $u_F \in M$ for the primitive inner normal, normalised by $\langle u_F, v\rangle = -1$ for $v \in F$; reflexivity is exactly the statement that u_F is a lattice point for every facet. $X \subset \mathbb{P}_\Delta$ denotes a Δ-regular anticanonical hypersurface in the sense of [Bat94, Def. 3.1.1]; it is a Calabi–Yau threefold with at worst Gorenstein canonical singularities.

2.1. **The dictionary.** Let G be a 2-face of Δ and $G^* \subset \Delta^\circ$ its dual edge. Write $\ell(G^*)$ for the lattice length of G^*. A Δ-regular X meets the closed orbit of σ_G, a curve in $\mathbb{P}_\Delta$, in exactly $\ell(G^*)$ points, and near each of them

$$(X, x) \ \cong\ \big(\mathrm{Cone}(S_G),\, 0\big), \tag{3}$$

the affine cone over the toric surface $S_G := \mathrm{TV}(\text{face fan of } G)$ taken in the polarisation given by G itself. Concretely, $\mathrm{Cone}(S_G)$ is the affine toric threefold of the cone over $G \times \{1\} \subset N_{\mathbb{R}} \oplus \mathbb{R}$.

The germ (3) is a singular point of X precisely when G is not a unimodular triangle, and it is an *isolated* singular point precisely when G has no lattice points on the interiors of its edges, i.e. when every edge of G has lattice length one. We will only meet the following three:

G	S_G	germ
unit square	$\mathbb{P}^1 \times \mathbb{P}^1$	ordinary double point (node), $xy = zw$
pentagon, one interior point	$\mathrm{dP}_7 = \mathrm{Bl}_2\mathbb{P}^2$	$\mathrm{Cone}(\mathrm{dP}_7)$
hexagon, one interior point	$\mathrm{dP}_6 = \mathrm{Bl}_3\mathbb{P}^2$	$\mathrm{Cone}(\mathrm{dP}_6)$

A pentagon with exactly one interior lattice point is reflexive after translating that point to the origin, and its face fan is the fan of dP_7; this is the sense in which the second row is a del Pezzo cone.

Definition 2.4. Δ lies in the *admissible framework* if every 2-face of Δ is either a unimodular triangle, or a unit square, or a pentagon or hexagon with one interior point; every edge of Δ has lattice length one; and every singular 2-face has $\ell(G^*) = 1$. For such a Δ, $\mathrm{Sing}(X)$ is a finite set with one point per singular 2-face, and every germ is a node or an anticanonical del Pezzo cone of degree 6 or 7.

This is the class in which [Wue26b] proves its criterion, and the class we work in throughout. All three germs are smoothable: the node by the Milnor fibre of $xy = zw$, and $\mathrm{Cone}(\mathrm{dP}_d)$ for $d = 6, 7$ because a del Pezzo surface of degree ≥ 3 is projectively normal and its anticanonical cone deforms to the smooth generic hyperplane section; see [Alt97] for the versal base.

2.2. **Nodes, coloops, and when the nodal mechanism is unavailable.** Let X have nodes $x_1, \ldots, x_k$, and let $\widehat{X} \to X$ be a small resolution of the nodes with exceptional curves

$C_1, \dots, C_k$. Following [Wue26b, §5.2] we call the matroid of the vectors $[C_j]$ the *node relation system*, and a node x_j a *coloop* when $[C_j]$ lies in no circuit; equivalently, when every relation $\sum \lambda_i [C_i] = 0$ has $\lambda_j = 0$.

Remark 2.5. If some node is a coloop, no relation of the form (1) exists, and Friedman's criterion is vacuous: the nodal mechanism cannot smooth X by itself. The threefolds of interest here are exactly those where this happens and a del Pezzo cone point is also present, so that a smoothing, if it exists, must move both kinds of germ at once. We call such an X *genuinely mixed*.

Both X_{19} and X_9 are genuinely mixed: the node system of X_{19} has rank 12 with two coloops, and that of X_9 has rank 2 with both nodes coloops.

3. Deformations of the ambient by Minkowski decomposition

We now recall the mechanism. The reader who knows [IV12] can skim to Definition 3.3, which fixes the object the two theorems quantify over.

3.1. **Altmann's construction, affine case.** Let $\widetilde{N}$ be the lattice of the germ, unrelated to the ambient N of Convention 2.3, and $\widetilde{M}$ its dual. Let $\delta \subset \widetilde{N}_{\mathbb{R}}$ be a pointed cone and $r \in \widetilde{M}$ a degree. Altmann [Alt95, Alt97] showed that Minkowski decompositions

$$\delta_r := \delta \cap [r = 1] = \Delta_0 + \cdots + \Delta_l$$

into polyhedra with common tail cone $\sigma := \delta \cap [r = 0]$ give l-parameter deformations of the affine toric variety $\mathrm{TV}(\delta)$, with deformation parameters in degree $-r$. For an isolated Gorenstein toric threefold germ, T^1 is concentrated in the single degree $-R^*$ and has dimension $\#(\text{vertices}) - 3$ [Alt95, Thm. 6.5]; for the pentagon this is 2, and the versal base is a line with an embedded component [Alt95, (7.1.4)]. The pentagon has, up to translation, exactly one nontrivial decomposition into lattice polyhedra, namely

$$\text{pentagon} \;=\; \text{primitive segment} \;+\; \text{unimodular triangle}, \tag{4}$$

and the corresponding one-parameter deformation smooths $\mathrm{Cone}(\mathrm{dP}_7)$, because the cones over a primitive segment and over a unimodular triangle are both smooth.

3.2. **The non-affine case: complexity-one T-varieties.** Our ambient $\mathbb{P}_\Delta$ is complete and, for the polytopes we need, not simplicial, so the affine statement does not apply directly. Ilten–Vollmert [IV12] give the required globalisation, in the language of polyhedral divisors of Altmann–Hausen [AH06, AHS08].

Fix a primitive degree $R \in M$ and *downgrade* the torus action of $\mathbb{P}_\Delta$ along R: restrict from the full four-torus to the three-dimensional subtorus $\ker R$. By [IV12, Rem. 1.8] the resulting complexity-one T-variety is described by a divisorial fan on $\mathbb{P}^1$ whose two nontrivial *slices* are the cross-sections

$$\mathcal{S}^+ := \Sigma \cap [R = 1], \qquad \mathcal{S}^- := \Sigma \cap [R = -1], \tag{5}$$

polyhedral complexes in the rank-three lattice $R^\perp \cap N$ together with the tail fan $\Sigma \cap [R = 0]$. The maximal cells of $\mathcal{S}^\pm$ are the cross-sections $\sigma_F \cap [R = \pm 1]$ of the maximal cones; a cell is *bounded* exactly when R is positive (resp. negative) on every ray of σ_F: a point of $\sigma_F \cap [R = 1]$ is $\sum_i \lambda_i v_i$ with $\lambda_i \geq 0$ and $\sum_i \lambda_i \langle R, v_i \rangle = 1$, so each $\lambda_i \leq \langle R, v_i \rangle^{-1}$ when all $\langle R, v_i \rangle > 0$, and the cell is empty when all are negative. We call the union $\mathcal{S} := \mathcal{S}^+ \sqcup \mathcal{S}^-$ the *slice complex* at R.

Definition 3.1 ([IV12, Def. 2.1, 2.2, 4.1])**.** A *Minkowski decomposition of the slice complex* assigns, separately on each of the subdivisions $\mathcal{S}^+$ and $\mathcal{S}^-$, to every cell C a decomposition $C = C^0 + C^1$ into polyhedra with the same tail cone as C, subject to

- (D1) both summands carry the full tail cone, and $C^0 + C^1 = C$;
- (D2) *admissibility*: for each lattice character in the dual of the tail cone, at most one of the corresponding faces of the summands fails to contain a lattice point;
- (D3) if $C \cap D \neq \varnothing$, then $(C \cap D)^i = C^i \cap D^i$;
- (D4) for nonempty collections of cells $\mathcal{J} \subseteq \mathcal{I}$ in the same subdivision and every $A \subseteq \{0, 1\}$,

$$\sum_{i\in A} \bigcap_{C\in\mathcal{I}} C^i \prec \sum_{i\in A} \bigcap_{C\in\mathcal{J}} C^i.$$

The empty polyhedron is allowed as a face; the empty index set A is tautological.

The pairwise consequence of (D4) is the one that does the work in Theorem 5.2: take $\mathcal{J} = \{\lambda\}$, $\mathcal{I} = \{\kappa, \lambda\}$ and $A = \{i\}$ when $\kappa \prec \lambda$. Together with (D3), this says that the decomposition induced on a face is a face of the decomposition of the cell above it. In particular, if a cell admits only homothetic summands, then so does every face of it.

Theorem 3.2 ([IV12, Thm. 4.4])**.** *A Minkowski decomposition of the slice complex in the sense of Definition 3.1 determines a flat family $\pi\colon \mathcal{P} \to B$ over a curve germ B, with $\mathcal{P}_0 = \mathbb{P}_\Delta$ and with T-action on every fibre.*

Definition 3.3. The family of Theorem 3.2, for a fixed primitive $R \in M$ and a fixed decomposition of the slice complex at R, is the *single-degree family at R*. Its fibre over $b \in B$ is denoted $\mathcal{P}_b$, and the base parameter is written b throughout, the letter t being reserved for edge dilations.

Theorem 3.4 ([IV12, Cor. 2.12])**.** *Suppose the polyhedral divisor at a germ has* affine locus*. Then the singularities of the general fibre at that germ are the cones over the summands C^0, C^1. In particular the germ is smoothed if and only if each summand cone is smooth; for the bounded summands used below, this means a point or a unimodular simplex, in particular a primitive segment in dimension one.*

Theorem 3.5 ([IV12, Rem. 2.13])**.** *If a polyhedral divisor has* complete locus *and its associated affine T-variety is singular, no T-deformation obtained by decomposing that polyhedral divisor is a smoothing. Thus the general fibre is singular; the remark does not assert that its analytic germ is unchanged.*

Two elementary consequences will let us pass honestly from singular ambient fibres to their anticanonical hypersurfaces.

Proposition 3.6 (The relative Fano package)**.** *Let $\pi\colon \mathcal{P} \to B$ be a single-degree family of $\mathbb{P}_\Delta$, with Δ reflexive and B a curve germ. After shrinking B, the morphism π is proper and Gorenstein of relative dimension 4, and*

$$\mathcal{L} := \omega_{\mathcal{P}/B}^{-1}$$

is π-ample and satisfies $\mathcal{L}|_{\mathcal{P}_b} \cong \omega_{\mathcal{P}_b}^{-1}$ for every $b \in B$.

Proof. The divisorial fan of the special fibre is complete, so [HI11, Thm. 3.2] makes π proper; flatness is Theorem 3.2. The special fibre $\mathbb{P}_\Delta$ is Gorenstein. The fibrewise Gorenstein locus of a flat morphism of finite presentation is open [Stacks, Tag 0C09]; it contains the whole proper fibre $\mathcal{P}_0$, so properness permits us to shrink B until every fibre is Gorenstein. Thus π

is Gorenstein, and relative duality gives the asserted invertible dualizing sheaf and its base-change isomorphisms [Con00]. Finally, $\mathcal{L}_0 \cong \mathcal{O}_{\mathbb{P}_\Delta}(-K_{\mathbb{P}_\Delta})$ is ample because Δ is reflexive. Relative ampleness is open for a line bundle under a proper morphism, so another shrinking makes $\mathcal{L}$ relatively ample [Stacks, Tag 0D2S]. $\square$

Lemma 3.7 (A singular curve traps every ample Cartier section)**.** *Let Z be a normal projective variety, let L be ample, and suppose* $\operatorname{Sing}(Z)$ *contains a complete curve C. Every effective Cartier divisor $D \in |L|$ is singular.*

Proof. Pass to the normalization $\widetilde{C}$ of an irreducible component of C. If the section defining D vanishes identically on C, then D contains C. Otherwise it restricts to a nonzero section of the positive-degree line bundle $L|_{\widetilde{C}}$, and hence has a zero. In either case choose a closed point $x \in C \cap D$.

Put $A = \mathcal{O}_{Z,x}$ and let $g \in A$ define D. Since D is Cartier, g is a nonzerodivisor. If $A/(g)$ were regular, then A would be regular: indeed, if $g \notin \mathfrak{m}_A^2$, both embedding dimension and dimension rise by one from $A/(g)$ to A; while if $g \in \mathfrak{m}_A^2$, regularity of $A/(g)$ would give $\operatorname{edim} A = \dim(A/(g)) = \dim A - 1$, impossible. This contradicts $x \in \operatorname{Sing}(Z)$, so D is singular at x. $\square$

For a cone σ_G in the face fan the dichotomy is combinatorial. The locus is affine when $\langle R, \cdot\rangle$ is nonnegative or nonpositive on the rays of σ_G (we then say σ_G is *one-sided* at R), and complete when σ_G has rays strictly on both sides of $R^\perp$. If R vanishes on the whole cone, neither slice acts on its chart. This is the content of the reachability lemma below.

4. Reachability and locking

Throughout this section Δ is in the admissible framework and G is a singular 2-face.

4.1. Reachability.

Lemma 4.1 (Reachability)**.** *Let $R \in M$ be primitive. If σ_G has rays strictly on both sides of $R^\perp$, then the corresponding affine chart of the general fibre of every single-degree family at R has positive-dimensional singular locus.*

Proof. The rays of σ_G are the vertices of G. If $\langle R, \cdot\rangle$ takes both signs on them, then in the downgrade along R the polyhedral divisor attached to σ_G has nonempty coefficients at both 0 and ∞, hence complete locus. By Definition 2.4 the germ is a node or a del Pezzo cone, in particular singular, so Theorem 3.5 applies.

It remains to justify the asserted dimension. The tail cone $\tau = \sigma_G \cap R^\perp$ has dimension 2 in the rank-three acting lattice $N \cap R^\perp$. Choose a nonzero character u annihilating τ. Both u and $-u$ lie in $\tau^\vee$. For the proper polyhedral divisor $\mathcal{D}_b$ of the general fibre, semiampleness gives $\deg \mathcal{D}_b(u), \deg \mathcal{D}_b(-u) \geq 0$. Coefficient by coefficient,

$$\min_{v \in (\mathcal{D}_b)_y} \langle u, v\rangle + \min_{v \in (\mathcal{D}_b)_y} \langle -u, v\rangle \leq 0,$$

so both degrees are zero, equality holds at every coefficient, and $\mathcal{D}_b(-u) = -\mathcal{D}_b(u)$. After multiplying u to clear denominators, this degree-zero semiample divisor on $\mathbb{P}^1$ is principal. The coordinate ring therefore contains a nonconstant homogeneous unit $f\chi^u$ and its inverse. No point can be fixed by the acting torus while this unit is nonzero there. The singular locus is torus-invariant, so every one of its nonempty orbits has positive dimension. $\square$

Corollary 4.2. *If for every primitive $R \in M$ some singular 2-face of Δ is two-sided at R, then no relative anticanonical effective Cartier divisor in any single-degree family has smooth general fibre.*

Proof. By Lemma 4.1, the singular locus of the general ambient fibre has a positive-dimensional irreducible component. Its closure in the projective fibre is a positive-dimensional projective subvariety contained in the singular locus, and hence contains a complete curve. Apply Proposition 3.6 and Lemma 3.7. $\square$

Remark 4.3. The quantifier in Corollary 4.2 must range over *all* primitive degrees, not over the degrees $R = u_F$ dual to a facet F. With the normalisation $\langle u_F, v\rangle = -1$ these are exactly the vertices of the polar polytope Δ°, and we call them *vertex degrees* for that reason; at such an R the cell of F is the unique maximal bounded cell of the level-(-1) slice, and is F itself, which is why we call it the *lattice facet*. On Δ_{19} the vertex degrees leave at most 13 of the 15 singular 2-faces one-sided, while a general primitive degree, for instance $R = (-3, 0, 0, 1)$, leaves all 15 one-sided. So reachability does *not* obstruct X_{19}; what obstructs it is rigidity, which is Theorem 5.2. Restricting to vertex degrees would have given the right conclusion for the wrong reason.

Remark 4.4. The lemma is not vacuous. There is a reflexive 4-polytope with twelve vertices whose hypersurface has twelve singular points and for which no primitive degree in the search range leaves more than 10 of the 12 singular 2-faces one-sided, so Corollary 4.2 rules it out.

4.2. **The local degree gate.** Fix a unit-edge pentagonal 2-face G and put

$$L_G = N \cap \operatorname{span}_{\mathbb{R}}(\sigma_G), \qquad Y_G = \mathrm{TV}_{L_G}(\sigma_G).$$

Then L_G is saturated of rank 3, Y_G is the isolated Gorenstein threefold cone over dP_7, and a choice of splitting gives

$$U_{\sigma_G} \cong Y_G \times \mathbb{G}_m.$$

Let $R^*_G \in L_G^\vee$ be the Gorenstein degree, characterized by $\langle R^*_G, v\rangle = 1$ on every ray of σ_G.

Definition 4.5. If $r \in M$ is nonnegative and nonzero on σ_G, its *crosscut* is

$$Q_G(r) := \sigma_G \cap [r = 1].$$

It is a polyhedron with tail $\sigma_G \cap r^\perp$. When r is positive on all five rays it is the rational polygon $\operatorname{conv}\{v/\langle r, v\rangle : v \in V(G)\}$.

Lemma 4.6 (Local degree gate)**.** *Let $R \in M$ be primitive and suppose σ_G is one-sided at R but $R|_{L_G} \neq 0$. Choose $\epsilon \in \{1, -1\}$ so that $r = \epsilon R$ is nonnegative on σ_G. For every single-degree family at R:*

1. *if $r|_{L_G} \neq R^*_G$, the induced deformation of U_{σ_G} is trivial;*
2. *if $r|_{L_G} = R^*_G$, then $\langle r, v\rangle = 1$ for all $v \in V(G)$ and every admissible induced decomposition of $G = Q_G(r)$ is a decomposition into lattice polygons. If that decomposition is trivial, the induced deformation of U_{σ_G} is trivial.*

Thus a one-sided single-degree family can move the transverse dP_7 cone only on the affine lattice

$$\mathcal{A}_G := \{r \in M : \langle r, v\rangle = 1 \text{ for every } v \in V(G)\}.$$

Proof. The deformation datum lives in the cone $\sigma_G \subset (L_G)_{\mathbb{R}}$. After choosing the splitting $U_{\sigma_G} = Y_G \times \mathbb{G}_m$, the restriction of the Ilten–Vollmert construction to this chart is the Laurent extension of the homogeneous Altmann deformation of Y_G associated with the same crosscut

[IV12, §3]. Its Kodaira–Spencer class in $T^1(Y_G)$ has degree $-r|_{L_G}$ [Alt95, (5.3)]. Altmann's theorem for an isolated three-dimensional Gorenstein toric singularity says that $T^1(Y_G)$ is concentrated in degree $-R^*_G$ [Alt95, Thm. (6.5)]. Hence the Kodaira–Spencer map is zero unless $r|_{L_G} = R^*_G$.

Zero Kodaira–Spencer class is enough here, although it is not enough for an arbitrary deformation. Altmann's classification of the transverse admissible deformation datum with zero Kodaira–Spencer map gives only types A and B [Alt95, Prop. (5.6)]. Type A yields a trivial family [Alt95, (5.7)]; and a nontrivial type B datum does not exist when the toric variety is smooth in codimension two [Alt95, (5.8)]. The transverse variety Y_G is smooth in codimension two because its singularity is isolated. Thus its Altmann family is trivial, and so is its Laurent extension to U_{σ_G}. This proves (1).

In the remaining degree the crosscut is G itself. Altmann's Gorenstein specialization says precisely that admissibility is equivalent to all summands being lattice polytopes [Alt95, Thm. (6.6)]. A trivial decomposition is type A, so (5.7) again makes the induced family trivial. □

4.3. **Locking: a combinatorial rigidity criterion.** The remaining ingredient is a way to see that a cell of the slice complex has only homothetic Minkowski summands. The obvious route is to compute the dimension of its Minkowski summand cone, and that is what the first version of Theorem 5.2 did, once per degree. It is the wrong route, and the reason locates the right one.

Definition 4.7. A polyhedron C with tail cone τ is *Minkowski-rigid* if every decomposition $C = C^0 + C^1$ with $\operatorname{tail} C^i = \tau$ has each C^i a homothet of C up to translation. Equivalently, the *Minkowski summand cone* of C has dimension one, this being the solution cone of the *edge-cycle system*: one variable $t_e \geq 0$ for each bounded edge, and for each bounded 2-face with edge vectors $e_1, \dots, e_k$ in cyclic order the *closing condition*

$$\sum_{i=1}^{k} t_i e_i = 0. \tag{6}$$

The all-ones vector always solves (6), since the edge vectors of a 2-face sum to zero; it is the homothety, and the Minkowski summand cone has dimension at least one.

Remark 4.8. The dimension of the Minkowski summand cone is *not* an invariant of the cone σ_F: it depends on the degree. On Δ_{19}, facet 12 has summand cone of dimension 2 at the lattice facet and dimension 1 at other cells of σ_{12}. So a criterion phrased in terms of that dimension cannot be degree-independent, and the quantifier over M would have to be re-run every time. What follows is a criterion that is degree-independent by construction.

The reason such a criterion exists is elementary.

Lemma 4.9 (Cross-section)**.** *Let $\sigma \subset N_{\mathbb{R}}$ be a pointed cone of dimension n and let $R \in \operatorname{int} \sigma^\vee$. Then $C_R := \sigma \cap [R = 1]$ is a polytope of dimension $n-1$, its faces are the cross-sections of the nonzero faces of σ, and its face lattice is that of σ with the apex removed. In particular, for any two $R, R' \in \operatorname{int} \sigma^\vee$ the polytopes C_R and $C_{R'}$ are combinatorially isomorphic, by the isomorphism matching cross-sections of the same face.*

Proof. R is positive on $\sigma \setminus \{0\}$, so every ray meets $[R = 1]$ in exactly one point and C_R is bounded; the correspondence $\sigma' \mapsto \sigma' \cap [R = 1]$ between nonzero faces of σ and faces of C_R is inclusion-preserving in both directions. □

The edge *directions* of C_R do change with R: the vertex of C_R on the ray through v is $v/\langle R, v\rangle$, so an edge from v to v' has direction $v'/\langle R, v'\rangle - v/\langle R, v\rangle$. The point is that the forcing rules below never use the directions, only the following two facts, both of which hold in every cross-section:

(a) the edge vectors of a bounded 2-face sum to zero around the cycle;
(b) two consecutive edges of a convex polygon are linearly independent.

Proposition 4.10 (The forcing rules)**.** *Let C be a polyhedron, f a bounded 2-face of C with edge vectors $e_1, \dots, e_k$ in cyclic order, and $t_1, \dots, t_k$ a solution of the edge-cycle system.*

(1) ***Triangle.*** *If $k = 3$ then $t_1 = t_2 = t_3$.*
(2) ***Single.*** *If $t_i = \lambda$ for every i except one, say $i = k$, then $t_k = \lambda$.*
(3) ***Adjacent.*** *If $t_i = \lambda$ for every i except two* consecutive *indices, say $i = 1, 2$, then $t_1 = t_2 = \lambda$.*

None of the three refers to the lattice, to the degree, or to the edge directions beyond (b).

Proof. (1) By (a), $e_3 = -e_1 - e_2$, so (6) reads $(t_1 - t_3)e_1 + (t_2 - t_3)e_2 = 0$, and e_1, e_2 are independent by (b). (2) By (a), $\sum_{i<k} e_i = -e_k$, so (6) reads $(t_k - \lambda)e_k = 0$ and $e_k \neq 0$. (3) By (a), $\sum_{i\geq 3} e_i = -(e_1 + e_2)$, so (6) reads $(t_1 - \lambda)e_1 + (t_2 - \lambda)e_2 = 0$, and e_1, e_2 are consecutive, hence independent by (b). □

Remark 4.11*.* The word *consecutive* in (3) is not decoration. Two *opposite* edges of a quadrilateral may be parallel, and then the same computation gives only one equation for two unknowns. A version of rule (3) firing on opposite pairs would report the cube rigid; the cube is the Minkowski sum of three segments and has Minkowski summand cone of dimension three.

Definition 4.12. Let F be a 3-polytope and S a set of vertices of F. Maintain a partition of the edge set of F, initially into singletons, and call two edges *tied* when they lie in a common class. Using only those 2-faces all of whose vertices lie in S, apply the following three rules until none applies.

(T) *Triangle.* Merge the three edges of a triangular 2-face into one class.
(S) *Single edge.* If all but one edge of a 2-face are tied to one another, merge the remaining edge into their class.
(A) *Adjacent pair.* If all but two edges of a 2-face are tied to one another and those two are consecutive, merge both into their class.

Each application either changes nothing or decreases the number of classes, so the propagation terminates. Its outcome does not depend on the order in which the rules are applied: each rule's hypothesis is preserved under coarsening the partition, so the three rules define a monotone closure operator and the terminal partition is its least fixed point. Say that a set E of edges is *locked in* (F, S) if at termination all of E lies in a single class; that a facet F is *locked* if its whole edge set is locked in $(F, V(F))$; and that a 2-face G lying in the facets F_1, F_2 is *locked* if both F_1 and F_2 are locked. This last is hypothesis (L) of Theorem 5.1, and it is what the census of Section 6 counts.

The requirement in (S) and (A) that the other edges be tied *to one another*, and not merely each already forced, is what makes the rules valid: what they propagate is the value of a common dilation parameter, and two edges carrying forced but unequal parameters license nothing. A partition, rather than a set of forced edges, is what records that.

The propagation needs a seed, and (T) is the only rule that fires from singletons, so a 3-polytope with no triangular 2-face is never locked.

Proposition 4.13. *Let F be a facet of Δ and $R \in M$ with $\langle R, \cdot\rangle > 0$ on every vertex of F. If F is locked then the cell $\sigma_F \cap [R = 1]$ is Minkowski-rigid.*

Proof. By Lemma 4.9 the cell is a polytope combinatorially isomorphic to F, with the same triangular 2-faces and the same edge-adjacency. By Proposition 4.10 every step of the propagation is a valid deduction about any solution of the edge-cycle system, so all t_e are equal and the Minkowski summand cone has dimension one. $\square$

Corollary 4.14. *Let G be a unit-edge pentagonal 2-face, let $r \in \mathcal{A}_G$, and let F be a facet containing G. If r is positive on every vertex of F and F is locked, then every decomposition of the slice complex induces the trivial lattice Minkowski decomposition on G. Consequently the induced deformation of U_{σ_G} is trivial.*

Proof. Proposition 4.13 makes $C = \sigma_F \cap [r = 1]$ Minkowski-rigid, so in a decomposition $C = C^0 + C^1$ we have $C^i = \lambda_i C + t_i$ with $\lambda_0 + \lambda_1 = 1$. Condition (D3) and the pairwise consequence of (D4) then give $G^i = \lambda_i G + t_i'$. Lemma 4.6 says that both G^i are lattice polytopes. Every edge of G is primitive, so the lattice length of a corresponding edge of G^i is λ_i; hence $\lambda_i \in \mathbb{Z}_{\geq 0}$. Their sum is one, so one summand is a translate of G and the other is a point. This is the trivial decomposition, and the last assertion is Lemma 4.6(2). $\square$

4.4. **The sign lemma.** The affine lattice $\mathcal{A}_G$ is a line. The following determines how the two facets containing G behave along it; it is a statement about reflexive polytopes and does not involve deformation theory.

Lemma 4.15 (Sign lemma)**.** *Let Δ be reflexive, G a 2-face, and F_1, F_2 the two facets containing G. The annihilator of* $\operatorname{span}(G)$ *in M has rank one; let d generate it, normalised so that $u_{F_1} - u_{F_2} = c\, d$ with $c > 0$. Then*

$$d(w) \leq -\tfrac{1}{c} < 0 \quad \text{for } w \in V(F_1) \smallsetminus V(G), \qquad d(w') \geq \tfrac{1}{c} > 0 \quad \text{for } w' \in V(F_2) \smallsetminus V(G).$$

Proof. $\operatorname{span}(G)$ is three-dimensional, so its annihilator has rank one. Both u_{F_1} and u_{F_2} take the value -1 on G, so their difference kills $\operatorname{span}(G)$ and is an integer multiple of d; this defines $c \neq 0$ and fixes the sign of d. For $w \in V(F_1) \smallsetminus V(G)$ we have $u_{F_1}(w) = -1$, and $w \notin F_2$ with Δ reflexive gives $u_{F_2}(w) \geq 0$, whence $c\, d(w) = -1 - u_{F_2}(w) \leq -1$. Symmetrically $c\, d(w') = u_{F_1}(w') + 1 \geq 1$. $\square$

Corollary 4.16. *Fix a pairing vector on G and let $R_\tau = R_0 + \tau\, d$ be the corresponding line of degrees. Then the cell of F_1 is bounded exactly for $\tau < T_1$ and the cell of F_2 exactly for $\tau > T_2$, for thresholds T_1, T_2 computed from R_0 and d by finitely many divisions. In particular the sign pattern of R_τ on $V(F_1) \cup V(F_2)$ takes at most $\#(V(F_1) \cup V(F_2)) + 1$ values along the line.*

5. The obstruction: a rigidity criterion, and X_{19}

We can now state the obstruction for a class of polytopes rather than for one.

Theorem 5.1. *Let Δ be a reflexive 4-polytope in the admissible framework and let G be a unit-edge pentagonal 2-face, with containing facets F_1, F_2. Suppose that*

- (L) *both F_1 and F_2 are locked;*
- (C) *for every $r \in \mathcal{A}_G$, the functional r is positive on every vertex of F_1 or on every vertex of F_2.*

Then, for every primitive $R \in M$, the general ambient fibre of every single-degree family at R has positive-dimensional singular locus. After shrinking the base, no relative anticanonical effective Cartier divisor in such a family has smooth general fibre.

The hypothesis is decidable: $\mathcal{A}_G$ is an affine lattice line, and condition (C) *is a finite comparison of the thresholds in Corollary 4.16.*

Proof. Let R be primitive. If σ_G is two-sided at R, Lemma 4.1 already gives a singular curve in the corresponding chart. If R vanishes on L_G, neither slice changes U_{σ_G}, so its singular locus remains $\mathrm{Sing}(Y_G) \times \mathbb{G}_m$.

It remains to consider the one-sided case with nonzero restriction. Choose $r = R$ or $-R$ nonnegative on σ_G. If $r|_{L_G} \neq R^*_G$, Lemma 4.6(1) makes the local family trivial. If $r|_{L_G} = R^*_G$, then $r \in \mathcal{A}_G$; by (C), one containing facet is positive at r, and by (L) it is locked. Corollary 4.14 again makes the local family trivial. Thus in every one-sided case the general fibre contains the singular curve $\mathrm{Sing}(Y_G) \times \mathbb{G}_m$ in this chart.

Its closure in the proper ambient fibre is a complete curve contained in the singular locus in the one-sided case. In the two-sided case, Lemma 4.1 gives a positive-dimensional singular subvariety; its projective closure contains a complete curve in the singular locus, as in Corollary 4.2. Proposition 3.6 and Lemma 3.7 now give the assertion about anticanonical Cartier divisors. ☐

5.1. X_{19}. Let

(7)
$$\begin{aligned}\Delta_{19} = \mathrm{conv}\big\{&(1,0,0,0),\ (0,1,0,0),\ (0,0,1,0),\ (0,0,-1,0),\ (0,-1,0,0),\ (1,-1,-1,0),\\ &(0,0,0,1),\ (-1,1,1,-1),\ (0,-1,0,1),\ (0,0,-1,1),\ (-1,1,0,-1),\ (-1,0,1,-1),\\ &(-1,0,1,0),\ (-1,1,0,0),\ (0,-1,-1,0),\ (0,-1,-1,1),\ (-1,0,0,-1),\\ &(-2,1,1,-1),\ (-1,0,0,1)\big\},\end{aligned}$$

and label the displayed vertices $v_0, \dots, v_{18}$ in that order. This is a reflexive polytope with 19 vertices, 27 facets and 64 edges, all of lattice length one. It lies in the admissible framework: its singular 2-faces are fourteen unit squares and one pentagon $P = \{v_{14}, v_{15}, v_{16}, v_{17}, v_{18}\}$ with one interior point, each with dual edge of length one. So X_{19} has fourteen nodes and one dP_7 cone point, all isolated. The pentagon lies in exactly two facets, which we call F_{21} and F_{24} after their zero-based position in the ordering the certificates use. Their supporting equations $\langle u, \cdot \rangle = 1$ and vertex sets are

$$\begin{aligned}F_{21}:\ u &= (-1,0,-1,0), \quad \{v_3, v_9, v_{10}, v_{13}, v_{14}, v_{15}, v_{16}, v_{17}, v_{18}\},\\ F_{24}:\ u &= (-1,-1,0,0), \quad \{v_4, v_8, v_{11}, v_{12}, v_{14}, v_{15}, v_{16}, v_{17}, v_{18}\},\end{aligned}$$

and the facet F_{12} of Remark 4.8 is $u = (0,1,1,1)$ on $\{v_1, v_2, v_6, v_7, v_{12}, v_{13}, v_{17}, v_{18}\}$.

Theorem 5.2. Δ_{19} *satisfies the hypothesis of Theorem 5.1 at P. Consequently, for every primitive $R \in M$ the general fibre of every single-degree family of $\mathbb{P}_{\Delta_{19}}$ at R has a singular curve. No relative anticanonical effective Cartier divisor in such a single-degree ambient family, with central fibre X_{19}, has smooth general fibre.*

Proof. Both facets are locked: each has 9 vertices, 16 edges and nine 2-faces, of degrees $(3,3,3,3,3,4,4,4,5)$ summing to $32 = 2 \cdot 16$, and the propagation of Definition 4.12 closes on all 16 edges.

It remains to check (C), and here there is no enumeration. Solving $\langle r, v_i\rangle = 1$ for $i = 14, \dots, 18$ gives exactly

$$\mathcal{A}_P = \{r_s = (-1, s, -1-s, 0) : s \in \mathbb{Z}\}.$$

The five pentagon vertices pair to 1. Each of the other four vertices of F_{21} pairs to $s+1$, and each of the other four vertices of F_{24} pairs to $-s$. Hence r_s is positive on every vertex of F_{21} when $s \geq 0$ and on every vertex of F_{24} when $s \leq -1$. These two ranges cover every integer s, proving (C). Theorem 5.1 completes the proof. □

Remark 5.3. Theorem 5.2 is not a non-smoothability statement. It closes one mechanism. Multi-degree families are not covered, and Altmann's construction does allow $\delta_r = \Delta_0 + \cdots + \Delta_l$ with $l > 1$, which [IV12, §6] globalises; nor is any deformation not induced from the ambient. Since the criterion of [Wue26b] is silent on X_{19}, the expected answer is that X_{19} is smoothable; see Section 8.

Remark 5.4. The affine line in the proof really is the complete list of one-sided degrees that can move the transverse cone. The reason is not an a priori restriction on R: it is Altmann's concentration of T^1 in the Gorenstein degree, together with his classification of zero Kodaira–Spencer deformation data.

An earlier proof instead enumerated rational crosscuts whose summand cones contained smooth cones. That test omitted admissibility (D2), so it produced false positives and did not prove necessity. Neither that enumeration nor its area cutoff is used in Theorems 5.1 or 5.2.

6. A census: how special is the obstruction?

The forcing argument behind Theorem 5.2 is purely combinatorial, so it runs over the Kreuzer–Skarke classification [KS00]. Doing so separates a general obstruction from an accident of Δ_{19}, and the answer is the first: locking holds at a little over three quarters of the pentagonal 2-faces in the range. What is rare is not the obstruction but its failure, and the polytope of Theorem 7.12 is the only place in the range where it fails inside the framework.

6.1. **The sweep.** Locking is a property of the face lattice, so it can be decided for every reflexive polytope in the database. For each polytope, for each 2-face that is a pentagon with unit edges and one interior point, and for the two facets containing it, we run the propagation of Definition 4.12 and, when it stalls, compute the Minkowski summand cone of the lattice facet outright.

Each pentagonal 2-face is sorted into exactly one of three verdicts, in this order of precedence: *locked*, in the sense Definition 4.12 gives for a 2-face, that is, both containing facets have their whole edge sets in one class; *stalled*, if it does not, but the Minkowski summand cone of the lattice facet, computed outright, is one-dimensional, so that the cell is rigid without the chain certifying it; and *decomposing* otherwise, when a containing facet admits a nontrivial decomposition. The precedence makes the three exclusive.

The three verdict columns exhaust the pentagons: $8\,673 + 37 + 2\,886 = 11\,596$ at nine vertices, $9\,466 + 37 + 3\,005 = 12\,508$ in total. The sweep also reports a fourth verdict, for a configuration whose incidence data the test cannot read, and that class is empty at every vertex count, so the stalls are a limitation of what the chain certifies and not of the implementation.

Nothing in Theorem 5.1 or Theorem 6.3 depends on the 37: the hypothesis of both is locking, and the 77 framework polytopes are counted separately. What the middle column

TABLE 1. The locking sweep over the Kreuzer–Skarke classification. Each pentagonal 2-face is counted once, under the first verdict that applies.

vertices	polytopes	pentagons	locked	stalled	decomposing
5	1 561	0	n/a	n/a	n/a
6	24 189	0	n/a	n/a	n/a
7	177 446	30	30	0	0
8	834 638	882	763	0	119
9	2 867 955	11 596	8 673	37	2 886
total	3 905 789	12 508	9 466	37	3 005

changes is the answer to a different question. A reader who wants to know how many pentagons cannot be moved, rather than how many the chain proves cannot be moved, should read $9\,466 + 37 = 9\,503$.

So 9 466 of the 12 508 pentagonal 2-faces occurring at up to nine vertices are locked, that is, satisfy hypothesis (L) of Theorem 5.1, a little over three quarters of them, and the cell is rigid at 9 503. The sweep measures hypothesis (L) only; it does not test the affine-line coverage condition (C). Locking is common but not automatic: at nine vertices 2 886 pentagons, just under a quarter, sit in a facet that decomposes, and there Theorem 5.1 has nothing to say. What obstructs X_{19} is a feature of how its pentagon is glued into its two facets, and it is a feature most pentagons share.

Remark 6.1. The partition in Definition 4.12 is not a convenience. A procedure that recorded only which edges are forced, without recording which are forced *together*, would apply rules (S) and (A) when the other edges carry different constants, and would over-report locking; at nine vertices it gives 2 350 decomposing facets where the correct count is 2 886.

A decomposing facet is necessary but not sufficient, because by Definition 3.1(D4) what matters is the decomposition *induced* on the pentagon, and by (4) that must be segment + triangle with all dilation parameters nonnegative across the facet. Of the 119 at eight vertices: all 119 induce a nontrivial decomposition of the pentagon, 49 attain the segment-plus-triangle datum with nonnegative dilations, and 0 lie in a polytope whose whole germ inventory lies in the framework of Definition 2.4.

Remark 6.2. A filter that classified a 2-face by its numbers of vertices and interior points alone would return two admissible polytopes at eight vertices instead of none, so it will not do. A triangle with no interior point need not be smooth: it is smooth only if *unimodular*, and one with an edge of lattice length two gives a non-isolated germ. Exactly two of the 49 at eight vertices carry such a face.

6.2. **Nine vertices, and Δ_9.** At nine vertices the database contains 11 596 pentagonal 2-faces. Of these, 63 lie in a polytope inside the admissible framework, and 2 886 lie in a facet that decomposes. Exactly one does both, and it is the pentagon of Δ_9. That is how the polytope of Theorem 7.12 was found: the search was not for a smoothing but for the two combinatorial preconditions, and the smoothing is what the unique survivor turned out to admit.

Across all five vertex counts the number of pentagons in a framework polytope is 77, and the number lying in a decomposing facet is 3 005; the single polytope in both is Δ_9.

That uniqueness is what makes the example non-negotiable, and it also bounds what the construction of Section 7 can be expected to generalise to at this size.

6.3. **The searched range, classified.** Restricting the sweep to polytopes that lie in the admissible framework and carry a pentagonal 2-face returns exactly 77 of them, one pentagon apiece: one at seven vertices, thirteen at eight, sixty-three at nine. Running both theorems over all 77 decides every one.

Theorem 6.3. *Let Δ be a reflexive 4-polytope with at most nine vertices lying in the admissible framework and carrying a pentagonal 2-face. There are 77 such Δ, and:*

(1) *for 76 of them the criterion of* [Wue26b] *obstructs, so X admits no one-parameter smoothing* [Wue26b, Prop. 9.2] *and hence, by* [Wue26b, Cor. 6.7]*, is* not smoothable *over any base germ; 67 of these have a single singular point, always a lone dP_7 cone point, and 9 have one node in addition;*
(2) *for the remaining one, which is Δ_9, the criterion is silent, and X_9* is *smoothable by Theorem 7.12;*
(3) *exactly 76 of the 77 pentagons are locked, and the unlocked one is Δ_9's.*

Proof. The range is enumerated by `dp7_facet_sweep.py` and the count of 77 recorded there. Part (1) is [Wue26b, Prop. 9.2] evaluated on each, and the evaluation is `classification.py`, which also splits the 76 as $67 + 9$ by singular locus; the passage from "no one-parameter smoothing" to "not smoothable over any base germ" is [Wue26b, Cor. 6.7]. Part (2) is Proposition 7.2 for silence and Theorem 7.12 for smoothability. Part (3) is Definition 4.12 evaluated on the 77 pentagons by `locking.py`. $\square$

Remark 6.4. Part (1) of Theorem 6.3 is not proved here. It is the criterion of [Wue26b] run over the range, and it is stated as [Wue26b, Prop. 9.2]; we quote it because it is what makes the second part a classification rather than an example. It is new in bulk: [Wue26b] exhibits one non-smoothable Calabi–Yau threefold by hand, and the sweep produces seventy-six more, of which sixty-seven have a *single* singular point. The reason so many are obstructed is structural rather than accidental: with one or two germs there is too little for a relation to cancel against, and the criterion is a statement about cancellation.

Part (2) is the only case in the range where sufficiency is testable at all, and there it holds. So Conjecture 8.1 is consistent with all 77 and confirmed on the one that tests it.

Part (3) says the two mechanisms agree: in this range the ambient construction is available precisely where the criterion permits a smoothing, and nowhere else. We do not know a reason for that beyond the computation, and it is not forced by anything proved here: Theorem 5.2 shows the two can come apart, since X_{19} is locked with the criterion silent. Δ_{19} has nineteen vertices and so lies outside the swept range.

7. The construction: Theorem B

7.1. **The polytope and its germs.** Let Δ_9 be as in (2), with vertices

$$\begin{aligned} &v_0 = (1,0,0,0), && v_1 = (0,1,0,0), && v_2 = (-1,-1,0,0),\\ &v_3 = (0,0,1,0), && v_4 = (0,0,0,1), && v_5 = (-4,-2,-1,0),\\ &v_6 = (-4,-2,0,-1), && v_7 = (3,1,1,1), && v_8 = (2,1,1,1). \end{aligned}$$

It is reflexive, with 12 facets and 27 two-faces, and it lies in the admissible framework.

Proposition 7.1. $\mathrm{Sing}(X_9)$ *consists of exactly three points: two ordinary double points, at the* 2*-faces* $\{v_2, v_3, v_6, v_7\}$ *and* $\{v_2, v_4, v_5, v_7\}$*, and one* dP_7 *cone point, at the pentagon* $\{v_3, v_4, v_5, v_6, v_8\}$.

Proof. Every vertex of Δ_9 is primitive, so every ray of Σ is a smooth cone. Every edge has lattice length one and unit greatest common divisor of its 2×2 minors, so every two-dimensional cone is smooth; hence $\mathbb{P}_{\Delta_9}$ is smooth along the orbits of all cones of dimension ≤ 2. Of the 27 two-faces exactly the three listed have a singular cone, and each has dual edge of lattice length one, so contributes exactly one point. Finally, $\#(\Delta_9^\circ \cap M) = 162$ and this set contains every vertex of Δ_9°, so a generic anticanonical section is nonzero at every torus-fixed point and X_9 misses the zero-dimensional strata. ☐

Proposition 7.2. *The node relation system of* X_9 *has rank* 2 *and both nodes are coloops. The criterion of* [Wue26b] *is silent on* X_9*: at the* dP_7 *point the branch subspace is a single line, since* $K^\perp$ *has discriminant* 7 *and carries a unique* (-2)*-class up to sign, so the branch restriction there is one linear condition; the branch-restricted kernel is one-dimensional, and no coordinate vanishes identically on it.*

Proof. Both statements are computations on Δ_9, run in the certificates `v09_candidate.py` and `sing_locus.py`, and for the kernel in `relation_class.py` of [Wue26b]. The relation class $(-2, -2, 2)$, with the entries indexed by the two nodes and the (-2)-line at the dP_7 point, is nonzero at all three germs, so by [Wue26b, Rem. 6.2] no coordinate is forced. ☐

So X_9 is genuinely mixed in the sense of Remark 2.5, and the necessity criterion does not decide it. It is precisely an instance where the sufficiency question is open, and we now settle it.

7.2. **The family.** Take the vertex degree

$$R \;=\; u_{11} \;=\; (0, 1, -1, -1), \qquad \langle R, v\rangle = -1 \text{ for every } v \in F_{11},$$

dual to the facet $F_{11} = \{v_2, v_3, v_4, v_5, v_6, v_7, v_8\}$, which contains all three singular 2-faces. Downgrading along R, the bounded cell of the level-(-1) slice is the lattice facet F_{11} itself, so any Minkowski decomposition of it is into lattice polytopes and admissibility (Definition 3.1(D2)) is immediate.

The Minkowski summand cone of F_{11} has dimension 2. We take the decomposition given by the dilation vector

$$t = (0, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0)$$

on the eleven edges of F_{11}, listed in lexicographic order of their vertex index pairs as

$$(2, 5),\ (2, 6),\ (2, 7),\ (3, 6),\ (3, 7),\ (3, 8),\ (4, 5),\ (4, 7),\ (4, 8),\ (5, 6),\ (7, 8),$$

so the dilation is 1 on the three edges v_2v_7, v_3v_6, v_4v_5 and 0 on the other eight. Figure 1 shows F_{11}, its three singular 2-faces and the three dilated edges.

A dilation vector determines a summand through the *displacement maps* φ_0, φ_1 on the vertices of F_{11}: fix the base vertex v_2, set $\varphi_0(v_2) = 0$, and along each edge v_av_b put

$$\varphi_0(v_b) = \varphi_0(v_a) + t_{ab}\,(v_b - v_a),$$

which is well defined precisely because t solves the edge-cycle system, that being what makes the value independent of the path from v_2. Put $\varphi_1(v_a) = (v_a - v_2) - \varphi_0(v_a)$. Then $\operatorname{conv}\varphi_0(V(F_{11})) + \operatorname{conv}\varphi_1(V(F_{11})) = F_{11} - v_2$, so the two maps exhibit the summands, and the decomposition is extended cellwise over the slice complex: for a cell of the slice complex, written $\mathrm{cell}(G)$ for the cell over a face G of Δ, the summands are $\mathrm{cell}(G)^i =$

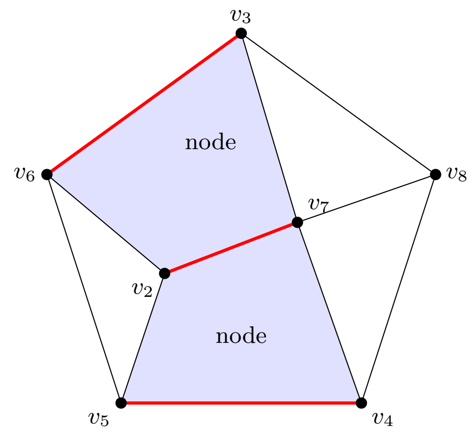

FIGURE 1. Schlegel diagram of the facet F_{11}, projected from beyond its pentagonal 2-face, which becomes the outer boundary. The two shaded faces are the squares $\{v_2, v_3, v_6, v_7\}$ and $\{v_2, v_4, v_5, v_7\}$ carrying the nodes; the third singular face is the pentagon $\{v_3, v_4, v_5, v_6, v_8\}$ carrying the dP_7 cone point, which the projection sends to the outer boundary. The three unshaded faces are triangles, and the germ there is smooth. The three heavy edges are those on which the dilation vector t is 1; it is 0 on the other eight.

$\operatorname{conv} \varphi_i(G \cap F_{11}) + \text{tail cell}(G)$, where φ_i is applied to whichever vertices of G lie in F_{11} and the empty convex hull is the origin. What this induces on the three singular 2-faces is:

TABLE 2. The decomposition induced at each germ.

2-face	germ	induced decomposition
$\{v_2, v_3, v_6, v_7\}$	node	primitive segment + primitive segment
$\{v_2, v_4, v_5, v_7\}$	node	primitive segment + primitive segment
$\{v_3, v_4, v_5, v_6, v_8\}$	Cone(dP_7)	primitive segment + unimodular triangle

Proposition 7.3. *The assignment above is a Minkowski decomposition of the two-sided slice complex in the sense of Definition 3.1. On the decomposed level-(-1) slice there are 69 nonempty cells and no failures of (D1) or (D2), and (D3) holds on every nonempty cell intersection. For the full arbitrary-collection axiom (D4), the summand-intersection semilattice has 86 states, and the required face relation holds on all 5 934 one-cell extensions and for all three nonempty subsets of $\{0, 1\}$. As pairwise diagnostics, these include all 346 face pairs and all 1 284 disjoint pairs, of which 1 028 have nonempty summand intersection even though the original cells do not meet. The undecomposed level-$(+1)$ slice has 29 nonempty cells; its trivial decomposition satisfies (D1)–(D4), with 29 intersection states and 841 one-cell extensions.*

Proof. Condition (D2) on the level-(-1) slice needs no computation: the unique maximal bounded cell is the lattice facet F_{11}, and φ_0, φ_1 send its vertices to lattice points because the dilations t_e are integers, so both summands are lattice polytopes and every lattice character takes integer values on them. The other three are finite checks: (D1) cell by cell over the 69 cells and (D3) on every nonempty pairwise intersection. For (D4), a state records the two intersections of summand cells over a nonempty collection. Adding one cell gives an inclusion of collections; checking that the three corresponding partial sums are faces on every such extension proves the axiom for arbitrary inclusions by transitivity of the face relation. This gives the 86 states and 5 934 extensions above. On the level-$(+1)$ slice we use $C = C + \mathrm{tail}(C)$. The tail summand contains the lattice point zero in every minimizing face, so (D2) is automatic there; the same exact semilattice check gives 29 states and 841 extensions. The two enumerations and all checks are `slice_admissible.sage`. □

Corollary 7.4. *By Theorem 3.2 the decomposition determines a flat family $\pi\colon \mathcal{P} \to B$ with $\mathcal{P}_0 = \mathbb{P}_{\Delta_9}$, and by Theorem 3.4 the general fibre $\mathcal{P}_b$ is smooth at all three germs of X_9: the cones over a primitive segment and over a unimodular simplex are smooth.*

All three cones σ_G are one-sided at R with bounded cells, so Lemma 4.1 does not intervene and the affine-locus hypothesis of Theorem 3.4 is satisfied at each.

Remark 7.5. The family is proper. This is *not* because π factors through $\mathbb{P}^1 \times B$. For a divisorial fan the map to the base curve is only rational [IV12, Lem. 1.6], and the map that does exist on affine charts is a good quotient, hence an affine morphism. The statement we use is Hochenegger–Ilten [HI11, Thm. 3.2]: $X(\mathfrak{S}^{\mathrm{tot}})$ is separated, and π is proper if and only if $\mathfrak{S}$ is complete. Ours is complete because the face fan of Δ_9 is. The same paper's [HI11, Thm. 5.1] applies cohomology and base change to this same family, which is the precedent for Section 7.3.

7.3. **The relative anticanonical system.** For the hypersurface to follow the ambient we need the anticanonical system to be constant in the family.

Proposition 7.6. *After shrinking the curve germ B, the proper flat morphism $\pi\colon \mathcal{P} \to B$ is Gorenstein of relative dimension 4. Its relative dualizing sheaf is therefore invertible and compatible with base change. In particular*

$$\mathcal{L} := \omega_{\mathcal{P}/B}^{-1} \qquad \textit{satisfies} \qquad \mathcal{L}|_{\mathcal{P}_b} \cong \omega_{\mathcal{P}_b}^{-1} = \mathcal{O}_{\mathcal{P}_b}(-K_{\mathcal{P}_b})$$

for every b near 0.

Proof. The special fibre $\mathcal{P}_0 = \mathbb{P}_{\Delta_9}$ is Gorenstein because Δ_9 is reflexive. For a flat morphism of finite presentation, the locus in the total space where the fibre is Gorenstein is open [Stacks, Tag 0C09]. This open set contains the whole proper fibre $\mathcal{P}_0$. The image of its closed complement is closed because π is proper and does not contain 0; deleting that image from B makes every fibre Gorenstein. Thus π is a Gorenstein morphism. Relative duality now gives an invertible $\omega_{\mathcal{P}/B}$ whose formation commutes with base change [Con00]. □

Proposition 7.7. $h^0(\mathcal{P}_b, \mathcal{L}_b) = 162 = h^0(\mathcal{P}_0, \mathcal{L}_0)$. *Consequently $\pi_*\mathcal{L}$ is locally free and $H^0(\mathcal{P}, \mathcal{L}) \to H^0(\mathcal{P}_b, \mathcal{L}_b)$ is surjective for all b near 0.*

The computation is not the toric one, because the general fibre is not toric:

Proposition 7.8. *$\mathcal{P}_b$ is not toric.*

Proof. By Süß [Sü08, Cor. 1.9] a complexity-one T-variety over $\mathbb{P}^1$ is toric if and only if all but at most two of its slices are translates of the tail fan by a lattice element. The general fibre has three nontrivial slices, and 43 of the 72 cells of those slices are empty, so no slice is such a translate. These are the cells of the general fibre's slices, not the 69 cells of the level-(-1) slice of Proposition 7.3. The slice data are computed in `toric_fibre.sage`. □

Remark 7.9. The translation in [Sü08, Cor. 1.9] is tightly constrained, and a looser reading would make the criterion apply. By [Sü08, Thm. 1.8] the translation at a point y is $\sum_i a_y^i e_i$ with the $D_i = \sum_y a_y^i y$ *principal divisors*; equivalently [AIPSV11, §5.3], all the translations are read off a single plurifunction $\sum_i v_i \otimes f_i \in N \otimes \mathbb{C}(Y)$. The whole slice at y moves by one vector, and the vectors at different points are not independent. No source permits different cells of the same slice to move differently.

So we use the formula of Petersen–Süß [PS11, Def. 3.22, Prop. 3.23] for h^0 of a divisor on a complexity-one T-variety. Its symbols are fixed first. A T-invariant divisor on such a variety is recorded by a *divisorial support function* h: for each point y of the base $\mathbb{P}^1$ a piecewise linear function h_y on the slice at y, together with the tail fan. Its Legendre dual h^* sends a character $u \in M$ to the divisor $h^*(u) = \sum_y \min_v\big(\langle u, v\rangle - h_y(v)\big)\, y$ on the base, the minimum taken over the vertices v of the slice at y, and the *weight box* $\Box_h \subset M_{\mathbb{R}}$ is the bounded polytope of u for which that divisor has degree at least -1. With these,

$$h^0(X, D_h) = \sum_{u \in \Box_h \cap M} \max\big(0,\ \deg\lfloor h^*(u)\rfloor + 1\big),$$

each lattice character in the box contributing the dimension of a linear system on $\mathbb{P}^1$. The canonical divisor is [PS11, Thm. 3.21], and its vertex values are [PS11, Cor. 3.19], $h_y(v) = K_Y(y) + 1 - 1/\mu(v)$, with $\mu(v)$ the least positive integer scaling v into the lattice. The linear part of h is then determined one maximal tail cone at a time, using the Cartier condition $\sum_y a_y(\sigma) = 0$, which says the translations at the several points cancel, and $h_y(n_\rho) = -1$ on the extremal rays n_ρ. We validate the implementation twice before use:

check	expected	obtained
[PS11, Ex. 3.25], $\mathbb{P}(\Omega_{\mathbb{P}^2})$	27 (printed there)	27
Δ_9 special fibre vs. $\#(\Delta_9^\circ \cap M)$	162	162

The second is a consistency check no source states as a theorem: the T-variety formula, evaluated over 89 weights on an eight-cone tail fan, reproduces a plain lattice-point count of the polar polytope. The general fibre's value is computed with three different degree-(-2) representatives of $K_{\mathbb{P}^1}$, all agreeing.

Proof of Proposition 7.7. The displayed formula gives 162 on the special fibre and on every nonzero fibre: the latter divisorial fans differ only by moving the third marked point of $\mathbb{P}^1$, and are isomorphic after rescaling the base coordinate. Upper semicontinuity therefore makes h^0 constant after shrinking B. Grauert's theorem gives local freedom of $\pi_*\mathcal{L}$ and base change. Shrinking once more, we may trivialise this vector bundle; evaluation of its constant sections then gives the asserted surjection onto every fibre. □

7.4. **Global generation.**

Proposition 7.10. *$\mathcal{L}_b \cong \mathcal{O}_{\mathcal{P}_b}(-K_{\mathcal{P}_b})$ is globally generated on $\mathcal{P}_0$ and on $\mathcal{P}_b$.*

Proof. On the toric special fibre, $\mathcal{L}_0 \cong \mathcal{O}_{\mathbb{P}_{\Delta_9}}(-K)$ is generated by its monomial sections: it is the ample Cartier line bundle associated with the reflexive polytope Δ_9. By Proposition 7.7, $E := \pi_*\mathcal{L}$ is locally free and formation of E commutes with base change. Consider the relative evaluation map

$$\mathrm{ev}\colon \pi^*E \longrightarrow \mathcal{L}.$$

Its restriction to $\mathcal{P}_0$ is the usual evaluation map $H^0(\mathcal{P}_0, \mathcal{L}_0) \otimes \mathcal{O}_{\mathcal{P}_0} \to \mathcal{L}_0$, hence is surjective. The support of $\mathrm{coker}(\mathrm{ev})$ is closed in $\mathcal{P}$ and misses the whole proper fibre $\mathcal{P}_0$. Since π is proper, its image in B is closed and misses 0. After deleting that image and shrinking the curve germ once more, ev is surjective on all of $\mathcal{P}$. Restricting to each fibre proves the assertion. □

7.5. **The general member.**

Proposition 7.11. $\mathrm{Sing}(\mathcal{P}_b)$ *is a finite set of torus-fixed points.*

Proof. We work through the twelve maximal charts, one per facet of Δ_9. Five of them have complete locus and seven affine locus. For the charts with affine locus, Süß [Sü08, Thm. 3.3] identifies the singularity analytically with the toric singularity of $\overline{\mathbb{Q}^+ \cdot (\{1\} \times \Delta_P)}$; for the charts with complete locus, [Sü08, Prop. 3.1] identifies the chart with the affine toric variety of $\delta = \overline{\mathbb{Q}^+ \cdot (\{1\} \times \Delta_P \cup \{-1\} \times \Delta_Q)}$, after absorbing the coefficients to at most two, which is legitimate because the hypothesis of that proposition is stated up to the equivalence of [Sü08, Thm. 1.7], one p-divisor at a time. On the general fibre this absorption leaves two coefficients per chart; on the special fibre, one.

Running this, no chart of $\mathcal{P}_b$ is singular in positive dimension. As a control, the same computation on $\mathcal{P}_0$ returns positive-dimensional singular loci in exactly the four charts that carry a singular 2-face, namely those of facets 5, 6, 8 and 11. □

Theorem 7.12. X_9 *is smoothable. Precisely, let* $g \in H^0(\mathcal{P}, \mathcal{L})$ *be general and* $\mathcal{X} := \{g = 0\}$*. Then* $\mathcal{X} \to B$ *is flat,* $\mathcal{X}_0$ *is a generic anticanonical hypersurface of* $\mathbb{P}_{\Delta_9}$*, and* $\mathcal{X}_b$ *is a smooth Calabi–Yau threefold for general* $b \in B$*.*

Proof. Use the trivialisation of $\pi_*\mathcal{L}$ from Proposition 7.7, and choose g from a general constant vector in its 162-dimensional fibre. A nonzero constant vector restricts to a nonzero section g_b on every fibre. The fibres $\mathcal{P}_b$ are integral normal T-varieties, so g_b is a nonzerodivisor. The fibrewise regular-section criterion therefore makes $\mathcal{X}$ a relative effective Cartier divisor: locally, multiplication by g gives an exact sequence

$$0 \longrightarrow \mathcal{L}^{-1} \xrightarrow{g} \mathcal{O}_{\mathcal{P}} \longrightarrow \mathcal{O}_{\mathcal{X}} \longrightarrow 0$$

which remains exact after restriction to every fibre. Since the first two terms are flat over B, the local criterion for flatness shows that $\mathcal{O}_{\mathcal{X}}$ is flat over B.

$\mathcal{X}_0$ is a general member of $|-K_{\mathbb{P}_{\Delta_9}}|$: by Proposition 7.7 the restriction $H^0(\mathcal{P}, \mathcal{L}) \to H^0(\mathcal{P}_0, \mathcal{L}_0)$ is surjective, so a general g restricts to a general section on the special fibre. Hence $\mathcal{X}_0$ has the singularities of Proposition 7.1.

Fix one $b_1 \neq 0$. By Proposition 7.10 the system $\mathcal{L}_{b_1}$ is globally generated, so Bertini makes its general member smooth away from $\mathrm{Sing}(\mathcal{P}_{b_1})$. Proposition 7.11 says that this singular locus is finite; global generation also makes a general section nonzero at each of its points. Thus smooth sections form a nonempty open subset of $H^0(\mathcal{P}_{b_1}, \mathcal{L}_{b_1})$. Under the trivialisation of $\pi_*\mathcal{L}$, its inverse image in the constant section space $V \cong \mathbb{C}^{162}$ is open and nonempty. The sections whose restriction to $\mathcal{P}_0$ is generic form another nonempty open subset of the same irreducible space V, so a general g has both properties. Smoothness is open in the flat family $\mathcal{X} \to B$; hence $\mathcal{X}_b$ is smooth for b in a nonempty open subset of B.

Finally relative adjunction gives

$$\omega_{\mathcal{X}/B} \cong (\omega_{\mathcal{P}/B} \otimes \mathcal{L})|_{\mathcal{X}} \cong \mathcal{O}_{\mathcal{X}}.$$

On the special fibre the sequence is

$$0 \longrightarrow \omega_{\mathcal{P}_0} \longrightarrow \mathcal{O}_{\mathcal{P}_0} \longrightarrow \mathcal{O}_{\mathcal{X}_0} \longrightarrow 0.$$

A complete toric variety has $H^i(\mathcal{P}_0, \mathcal{O}_{\mathcal{P}_0}) = 0$ for $i > 0$; Serre duality gives the corresponding vanishings for $\omega_{\mathcal{P}_0}$ below degree 4. The long exact sequence therefore gives $H^1(\mathcal{X}_0, \mathcal{O}_{\mathcal{X}_0}) = H^2(\mathcal{X}_0, \mathcal{O}_{\mathcal{X}_0}) = 0$. Upper semicontinuity for the proper flat family $\mathcal{X} \to B$ gives the same vanishings on nearby fibres. Thus every smooth good fibre has trivial canonical bundle and $h^1(\mathcal{O}) = h^2(\mathcal{O}) = 0$, so it is a Calabi–Yau threefold. □

Remark 7.13. The order of quantifiers in Theorem 7.12 is the substance of the argument. One does *not* fix the equation of a chosen X_9 and lift it: note that $W \supseteq H^0 \times \{0\}$, since *every* anticanonical member of the special fibre is singular, since each meets the three one-dimensional orbits over the singular 2-faces. So "lift a chosen equation and hope" proves nothing. One takes a general section of the total space and reads off both ends.

Remark 7.14. The deformation is small in a precise sense: of the cells of the slice complex, only seven change. Three are the cells of the singular 2-faces, where Table 2 and Corollary 7.4 show that the cones over the summands are smooth. The other four are higher-dimensional cells. Their orbit strata cannot be inferred from cell dimension in a complete-locus chart; their contribution to the singular locus is covered instead by the twelve-chart computation of Proposition 7.11.

7.6. **What Theorem B does and does not add.** It adds one confirmed instance of mixed sufficiency: a Calabi–Yau threefold whose nodes are all coloops, so that no nodal relation (1) exists, and which is nonetheless smoothable, by a family that moves the node germs and the del Pezzo cone germ simultaneously. It does not add a criterion. The polytope was found by search, and the census of Section 6 says the search space at nine vertices contained exactly one candidate.

8. The mixed Friedman criterion

The two theorems bracket a statement neither proves.

Conjecture 8.1 (Mixed sufficiency)**.** *Let X be a Calabi–Yau threefold whose singularities are isolated nodes and anticanonical dP_6 or dP_7 cone points, all locally smoothable, and let $\widehat{X} \to X$ be a crepant resolution. Suppose there is a class*

$$\kappa \in \ker\Big(\bigoplus_p H_2(L_p;\mathbb{Q}) \longrightarrow H_2(\widehat{X};\mathbb{Q})\Big), \qquad \kappa_p \in \mathcal{R}_p \setminus \{0\} \quad \textit{for every } p.$$

Then X is smoothable.

This is the exact converse of Theorem 2.1, and it specialises to Friedman's criterion when every germ is a node. It would subsume Theorem 7.12 as a corollary.

What makes it hard, concretely: Friedman's argument replaces the nodes by rational curves in a small resolution; the relation among their classes is what allows the local smoothings to be glued, and the deformation theory is controlled because the germs are hypersurface singularities with one-dimensional T^1. A dP_7 cone point has no small resolution, its crepant resolution being divisorial with an exceptional del Pezzo surface, so there is no exceptional curve to enter a relation, which is why [Wue26b] had to substitute the (-2)-line in $K^\perp$; and $\dim T^1 = 2$ with a versal base that is a line with an embedded component [Alt95, (7.1.4)], so the local theory is not the one-dimensional unobstructed picture the gluing assumes. A mixed sufficiency theorem needs a gluing argument that works with a divisorial exceptional locus. The closest existing work is Gross [Gro97].

8.1. **What the conjecture is, structurally.** Three facts about the criterion, all established in [Wue26b, §8], say what the conjecture is asking for.

The channels are the germ's T^1. For a singular 2-face G with vertices $v_1, \dots, v_n$ in cyclic order there are canonical isomorphisms

$$T^1_{\mathrm{TV}(\sigma_G)} \;\cong\; \Big\{a \in \mathbb{Q}^n \;:\; \textstyle\sum_j a_j v_j = 0\Big\} \;\cong\; K^\perp \subset \mathrm{Cl}(S_G)^*_{\mathbb{Q}},$$

all of rank $n-3$, under which the channel rows of [Wue26b] at G are a basis written in the ray coordinates [Wue26b, Prop. 8.2]; the first map is Altmann's edge dilations [Alt95, Thm. 6.5] sent to their jumps $a_j = t_j - t_{j-1}$ across the vertices. So the $K^\perp$ lattice that [Wue26b] introduced as a substitute for the missing exceptional curve class, and justified by what it made the period computation do, is canonically the tangent space to the germ's deformation functor. A ray relation is a curve class, with a_j the intersection number against the invariant divisor D_j; at a node the square's relation $v_a + v_c = v_b + v_d$ is the class of the exceptional curve of a small resolution, and the identification says the same reading survives at a del Pezzo cone point, where there is no such curve. Summing over germs, $\dim H^0(\mathcal{T}^1_X)$ is 4 for X_9 and 16 for X_{19}.

Concretely, at a pentagon G the channel block is a 2×5 matrix whose rows are a basis of the relations $\sum_j a_j v_j = 0$ among its vertices. At the pentagon of Δ_9, whose vertices in the order v_3, v_4, v_5, v_6, v_8 are

$$(0,0,1,0),\; (0,0,0,1),\; (-4,-2,-1,0),\; (-4,-2,0,-1),\; (2,1,1,1),$$

one such basis is

$$\begin{pmatrix} -1 & 1 & -1 & 1 & 0 \\ -1 & -2 & 1 & 0 & 2 \end{pmatrix},$$

as one checks by summing the columns. At a unit square v_a, v_b, v_c, v_d the block is the single row $(1,-1,1,-1)$, the class of the exceptional curve. The matrix of [Wue26b] is these blocks stacked, one block per germ, each expressed against the ambient divisor classes.

The computable kernel is the topological one. On the admissible framework the correction term $\sum_F \ell^*(F)(\ell(F^*)-1)$ over 2-faces of Δ in Batyrev's formula for $h^{1,1}$ vanishes term by term (a singular 2-face has dual edge of lattice length one by Definition 2.4 and a non-singular one is a unimodular triangle), so $h^{1,1}(\widehat{X})$ is purely toric, the ambient divisor classes span $H^{1,1}(\widehat{X};\mathbb{Q})$, and the matrix kernel certified in [Wue26b] equals the topological kernel $\ker\big(\bigoplus_p H_2(L_p;\mathbb{Q}) \to H_2(\widehat{X};\mathbb{Q})\big)$ [Wue26b, Prop. 8.3]. That paper needs only the containment of the second in the first, which is the safe direction for a necessity statement; Conjecture 8.1 needs the equality, since without it a sufficiency statement phrased in terms of the computable test would not be well posed. Note this is a different vanishing from the one in Proposition A.1: that one is the mirror correction and follows from the unit edges of Δ, this one from the framework condition on dual edges.

Forced coordinates and a single class are the same condition. A $\mathbb{Q}$-vector space is not a finite union of proper subspaces, so the hypothesis of Conjecture 8.1 is equivalent to no coordinate being forced to vanish. Within the admissible framework, where by the previous paragraph the computable kernel is the topological one, the conjecture therefore reads: X *is smoothable if and only if the criterion is silent*, with the two halves exactly complementary. Outside the framework the conjecture still makes sense, since it is stated topologically, but the computable test need not decide it.

8.2. **Consistency with the primitive case.** Gross classifies the contractions of a smooth Calabi–Yau threefold that cannot be factored, and proves that a primitive contraction of a divisor to a point has smoothable target unless the exceptional divisor is $\mathbb{P}^2$ or $\mathbb{F}_1$ [Gro97, Thm. 5.8]; degrees 6 and 7 are allowed. That looks like it could conflict with the criterion, since a threefold with a single del Pezzo cone point has nothing for a relation to cancel against. It does not, and the reason is the third of the facts recalled above. If a crepant resolution $\pi\colon \widehat{X} \to X$ has exactly one exceptional *divisor* E over a germ, contracted to a point, the other exceptional loci being curves, and X is $\mathbb{Q}$-factorial, then the channel rows at $\pi(E)$ vanish identically [Wue26b, Prop. 8.4]: every ambient class restricts to E inside the span of $E|_E = K_E$, hence pairs to zero against $K_E^{\perp}$. The hypothesis concerns the exceptional divisors, which lie over the del Pezzo cone points; a $\mathbb{Q}$-factorial X admits no projective crepant resolution with an exceptional curve at all, so the statement is about the divisorial germs and not about nodes.

That our examples are *not* $\mathbb{Q}$-factorial at their cone points is not new here, and is proved more directly in [Wue26a, Prop. C]: for a unit-edge Δ whose non-smooth 2-face has one interior lattice point and dual edge of length one, the contraction of the exceptional del Pezzo has relative Picard number $\rho(S) \in \{2, 3, 4\}$, so it factors and is never primitive, and X is not $\mathbb{Q}$-factorial at the point. What the present discussion adds is the converse reading, that the *rank* of the channel block measures that failure, and the reconciliation below.

So in the $\mathbb{Q}$-factorial case the kernel is everything, the criterion is silent, and Conjecture 8.1 predicts smoothability, which is [Gro97, Thm. 5.8]. Contrapositively [Wue26b, Cor. 8.5], a germ whose channel block has positive rank is a germ at which X fails to be $\mathbb{Q}$-factorial, and the relation kernel is a *defect* in the classical sense. On Δ_9 and Δ_{19} that block has rank 2, the maximum, so neither is $\mathbb{Q}$-factorial at its dP_7 point and [Gro97, Thm. 5.8] does not speak to either. This is also why Gross's machinery cannot reach these examples: the smoothing theorems of [Gro97] that could apply to a divisorial exceptional locus, namely Theorem 4.3 and Theorem 5.8, assume or derive $\mathbb{Q}$-factoriality. (His Theorem 3.8 does not, but it is restricted to complete intersection singularities, which a dP_d cone with $d > 4$ never is [Gro97, (3.3)].)

The concrete case is in [Wue26b, §9.2]: at seven vertices, the smallest count at which a pentagonal 2-face occurs at all, the framework contains exactly one polytope, and its hypersurface has a lone dP_7 cone point. Its channel block has rank 2, so the kernel is zero and the criterion certifies that threefold non-smoothable; and the same rank says it is not $\mathbb{Q}$-factorial there, so [Gro97, Thm. 5.8] never applied. The tension is resolved in an example, not only in principle.

8.3. **How much is already known.** The $\mathbb{Q}$-factorial case of Conjecture 8.1 is a theorem, and it covers the whole inventory of Definition 2.4.

Theorem 8.2 (Namikawa [Nam97])**.** *Let X be a Calabi–Yau variety having only isolated toric Gorenstein singularities. If X is $\mathbb{Q}$-factorial, then $\mathrm{Def}(X) \to \prod_P \mathrm{Def}(X, P)$ is surjective. In particular, if every germ is smoothable then X is deformed to a nonsingular Calabi–Yau variety.*

Nodes and dP_6, dP_7 cones are all isolated toric Gorenstein germs, and all three are smoothable, so the hypotheses reduce to $\mathbb{Q}$-factoriality alone. Note that the equality $\mu + \sigma = \dim T^1$ that appears in the literature around this result is not an extra hypothesis: it follows for toric germs from $\mu = 0$ together with Namikawa's inequality $\mu + \sigma \le \dim T^1 \le 2\mu + \sigma$.

By [Wue26b, Prop. 8.4] a $\mathbb{Q}$-factorial X has vanishing channel rows, so the criterion of [Wue26b] is silent there and Conjecture 8.1 predicts smoothability. Theorem 8.2 proves it. The two agree on the whole $\mathbb{Q}$-factorial locus, by different arguments, and in fact the local computations coincide. Namikawa attaches to a germ two invariants, which we write μ_p and σ_p to avoid the cones σ_F of this paper: μ_p counts the local complete intersection defect and σ_p the rank of the local class group modulo Picard. He computes the kernel at a germ to have dimension $\sigma_p + \mu_p$, which at a toric germ is $\sigma_p = \dim T^1$; all of T^1 dies, which is exactly the vanishing of the channel rows.

Corollary 8.3. *The open content of Conjecture 8.1 is the case where the criterion of* [Wue26b] *is silent and X is* not *$\mathbb{Q}$-factorial.*

That case is not empty and contains a confirmed instance. By [Wue26b, Cor. 8.5] both Δ_9 and Δ_{19} have channel blocks of rank 2 at their dP_7 points, so neither is $\mathbb{Q}$-factorial there; the criterion is silent on both; and X_9 is smoothable by Theorem 7.12 while X_{19} is undecided.

The corollary is not the reduction it may appear to be. Were X $\mathbb{Q}$-factorial, every ambient class would restrict to the exceptional divisor E inside the span of $E|_E = K_E$ and so pair to zero against $K^\perp$, and the block at that germ would vanish. But E is a toric surface whose boundary divisors are restrictions of the ambient toric divisors, so $\mathrm{Pic}(\mathbb{P}_{\widehat{\Sigma}})_{\mathbb{Q}} \to \mathrm{Pic}(E)_{\mathbb{Q}}$ is surjective and the block has the full rank $n-3$, which is positive at every singular 2-face. Hence no admissible X carrying one is $\mathbb{Q}$-factorial, a conclusion the census bears out at all 77 polytopes of Theorem 6.3 and at Δ_{19}, without resting on them. The $\mathbb{Q}$-factorial case is therefore consistent, and settled by Theorem 8.2, but it appears to be empty on the singular locus considered here; the open content of the conjecture is the singular part of the framework.

Remark 8.4. We have not read the printed [Nam97] directly; that volume of the journal is not digitised. There is an openly available announcement version, Max-Planck-Institut für Mathematik MPI/95-82, which we have read, but it is seven pages against the printed twenty-four, its numbering is not the printed numbering, and *it does not contain the statement above*: its main theorem requires the semi-universal deformation space of every germ to be smooth, with no toric alternative, and the word toric does not occur in it. That hypothesis excludes a dP_7 cone, whose versal base is a line with an embedded point. The statement above is Namikawa's own, quoted with his numbering from his survey [NamS96], and corroborated independently by Steenbrink [Ste97, p. 1376], who reports that the paper "admits smoothable isolated singularities which are either toric or have smooth semiuniversal base space". A stronger form, allowing each germ independently to be toric or to have nonsingular Kuranishi space, appears in Namikawa's 1996 Kinosaki symposium record. We flag also that [Nam97] is not the paper cited as `[Nam97]` in some of the literature, which is Namikawa's *Smoothing Fano* 3*-folds* in the same volume of the same journal.

8.4. **The sufficiency half of the conjecture.** Write $\mathbb{T}^1_X = \mathrm{Ext}^1(\Omega^1_X, \mathcal{O}_X)$. Since the germs here are not local complete intersections, one should check first that nothing is lost by computing with Ω^1_X rather than with the cotangent complex L_X. Nothing is.

Lemma 8.5. *Let X be a normal Gorenstein threefold whose non-lci locus is finite. Then*

$$\mathrm{Ext}^i(\Omega^1_X, \mathcal{O}_X) \;\cong\; \mathrm{Ext}^i(L_X, \mathcal{O}_X) \qquad (i \le 2),$$

both as sheaves and globally, compatibly with the local-to-global spectral sequence.

Proof. The cohomology sheaves $\mathcal{H}^{-q}(L_X)$ for $q \ge 1$ vanish where X is a local complete intersection, so they are supported on a finite set and have finite length. A finite-length module on a Cohen–Macaulay threefold has grade 3, so $\mathcal{E}xt^p(-,\mathcal{O}_X)$ and $\mathrm{Ext}^p(-,\mathcal{O}_X)$ vanish on it for $p<3$. In the spectral sequence $E_2^{p,q}=\mathrm{Ext}^p(\mathcal{H}^{-q}(L_X),\mathcal{O}_X)\Rightarrow \mathrm{Ext}^{p+q}(L_X,\mathcal{O}_X)$ a term with $q\geq 1$ contributing in total degree ≤ 2 has $p\leq 1<3$, hence vanishes. Only $q=0$ survives in degrees ≤ 2, and $\mathcal{H}^0(L_X)=\Omega^1_X$. □

So $\mathbb{T}^1_X$ and $\mathbb{T}^2_X$ are the deformation-theoretic Ext groups in the usual sense, and the obstruction map used below is the one they carry. Now let

$$0\to H^1(\Theta_X)\to \mathbb{T}^1_X \xrightarrow{\alpha} H^0(\mathcal{T}^1_X)=\bigoplus_p T^1_{X,p}\xrightarrow{\mathrm{ob}} H^2(\Theta_X)$$

be the low-degree exact sequence of the local-to-global Ext spectral sequence. At each singular point let $\mathrm{Sm}_p\subseteq T^1_{X,p}$ be the *smoothing locus*: the tangent cone to the smoothing components of the germ's versal base. By (4) and Altmann this is the whole line at a node, a line at a dP_7 point, and the union of the A_1 line with the A_2 plane at a dP_6 point; each component is smooth. The next two lemmas identify this deformation-theoretic locus with the subspace $\mathcal{R}_p$ of the link homology. This is the local input needed in the global argument.

Lemma 8.6 (local Hodge comparison)**.** *Let (Y,p) be a three-dimensional ordinary double point or the anticanonical cone over the toric del Pezzo surface E of degree 6 or 7. Let $\pi_p\colon \widehat{Y}_p\to Y$ be the small resolution at a node and $\widehat{Y}_p=\mathrm{Tot}(K_E)$ at a cone point, let Z_p be its exceptional locus, and put $U_p=\widehat{Y}_p\smallsetminus Z_p$. Then the connecting homomorphism gives an isomorphism*

$$T^1_{Y,p}\xrightarrow{\ \sim\ } K_p^{\mathrm{cr}}:=\ker\Big\{H^2_{Z_p}(\widehat{Y}_p;\Omega^2_{\widehat{Y}_p})\longrightarrow H^2(\widehat{Y}_p;\Omega^2_{\widehat{Y}_p})\Big\}. \tag{8}$$

There is also a natural isomorphism

$$H_2(L_p;\mathbb{C})\cong H^3(L_p;\mathbb{C})\xrightarrow{\ \sim\ } K_p^{\mathrm{cr}}, \tag{9}$$

and the two isomorphisms are compatible with the local Gysin map to any global crepant resolution containing $\widehat{Y}_p$.

Proof. The identifications $T^1_{Y,p}\cong H^1(U_p;\Theta_{U_p})\cong H^1(U_p;\Omega^2_{U_p})$ are Schlessinger's comparison and contraction with a local crepant volume form. The comparison applies to the non-lci cones: the hypothesis in [FL25, Lem. 1.17 and Lem. 4.1] is weakened in [FL25, Rem. 4.2] to an isolated singularity of depth at least 3.

For a node, $\widehat{Y}_p=\mathrm{Tot}_{\mathbb{P}^1}(\mathcal{O}(-1)\oplus\mathcal{O}(-1))$. The projection is affine, and the tangent sequence together with $H^1(\mathbb{P}^1,\mathcal{O}(m))=0$ for $m\geq -1$ gives $H^1(\widehat{Y}_p;\Theta_{\widehat{Y}_p})=0$. The local cohomology sequence therefore identifies the one-dimensional space $T^1_{Y,p}$ with K_p^{cr}. The residue calculation for the node identifies this class with the generator of $H^3(L_p)$; equivalently, this is the local Gysin identification in Friedman's construction [Fri86].

Now suppose $\widehat{Y}_p=\mathrm{Tot}(K_E)$, with projection $\rho\colon\widehat{Y}_p\to E$ and zero section again denoted E. Contraction with the crepant volume form identifies

$$\Omega^2_{\widehat{Y}_p}(\log E)(-E)\cong\Theta_{\widehat{Y}_p}(-\log E),$$

and the logarithmic tangent sequence is

$$0\longrightarrow\mathcal{O}_{\widehat{Y}_p}\longrightarrow\Theta_{\widehat{Y}_p}(-\log E)\longrightarrow\rho^*\Theta_E\longrightarrow 0.$$

Since ρ is affine and $\rho_*\mathcal{O}_{\widehat{Y}_p} = \bigoplus_{k\geq 0}\mathcal{O}_E(-kK_E)$, Kodaira vanishing and toric Bott vanishing give

$$H^1(\widehat{Y}_p;\mathcal{O}_{\widehat{Y}_p}) = \bigoplus_{k\geq 0} H^1(E;\mathcal{O}_E(-kK_E)) = 0,$$
$$H^1(\widehat{Y}_p;\rho^*\Theta_E) = \bigoplus_{k\geq 0} H^1\big(E;\Omega^1_E\otimes\mathcal{O}_E(-(k+1)K_E)\big) = 0.$$

Thus $H^1(\widehat{Y}_p;\Omega^2(\log E)(-E)) = 0$. The exact sequence of [FL25, Thm. 2.1(v)], whose theorem explicitly allows every three-dimensional isolated canonical Gorenstein germ, now gives (8). Moreover [FL25, Thm. 2.1(vi)] gives a natural inclusion

$$H^3(L_p;\mathbb{C}) = \mathrm{Gr}^2_F H^3(L_p;\mathbb{C}) \hookrightarrow K_p^{\mathrm{cr}}$$

and a decomposition of K_p^{cr} with this as one summand. If the boundary polygon has n vertices, Altmann's calculation gives $\dim T^1_{Y,p} = n-3$. The circle-bundle Gysin sequence gives $\dim H^3(L_p) = 9-\deg E = n-3$. Hence the other summand is zero and the displayed inclusion is an isomorphism. Its construction in local cohomology is compatible with the Gysin map, which proves the final assertion. ☐

Lemma 8.7 (the smoothing branches)**.** *Under the isomorphism $T^1_{Y,p}\cong H_2(L_p;\mathbb{C})$ of Lemma 8.6, the tangent cone Sm_p to each smoothing component is exactly the corresponding subspace $\mathcal{R}_p$: the whole line at a node, the unique root line at a dP_7 cone, and respectively the A_1 line and the A_2 plane at a dP_6 cone.*

Proof. Only the cone points require an identification. For a unit-edge polygon $P = (v_0,\dots,v_{n-1})$, put $e_i = v_{i+1}-v_i$. Altmann identifies the degree of the versal tangent space used here with

$$\mathcal{M}(P) = \left\{(t_i)\in\mathbb{C}^n : \sum_i t_ie_i = 0\right\}/\mathbb{C}(1,\dots,1), \tag{10}$$

and a lattice Minkowski decomposition gives the class of its edge-dilation vector (t_i). The Hodge isomorphism of Lemma 8.6 is natural under lattice automorphisms of P, with one necessary character: contraction with the crepant volume form twists equivariance by the determinant of the induced automorphism of the three-dimensional cone lattice.

There is one unit-edge reflexive pentagon up to $\mathrm{GL}_2(\mathbb{Z})$. Take

$$P_7 = \big((-1,-1),(0,-1),(1,0),(0,1),(-1,0)\big).$$

Its nontrivial decomposition has dilation vector $t=(0,1,0,1,0)$, up to replacing t by $1-t$, and spans the reduced versal base. The reflection $s(x,y)=(y,x)$ fixes $[t]\in\mathcal{M}(P_7)$ and has determinant -1. On $H_2(L_p)=K_E^{\perp}$ it acts by -1 on the unique root line and by $+1$ on the complementary line. The determinant twist therefore forces the Hodge image of $[t]$ to be the root line.

Likewise there is one unit-edge reflexive hexagon. Take

$$P_6 = \big((1,0),(1,1),(0,1),(-1,0),(-1,-1),(0,-1)\big).$$

The determinant-one rotation $r(x,y)=(x-y,x)$ cyclically permutes its vertices. On $\mathcal{M}(P_6)$ its characteristic polynomial is $(\lambda+1)(\lambda^2+\lambda+1)$. The (-1)-eigenspace is spanned by the alternating dilation $(0,1,0,1,0,1)$, the triangle-plus-triangle decomposition which gives Altmann's one-dimensional smoothing component; the complementary plane is spanned by the coarsenings of the three-segment decomposition and is the two-dimensional component.

On $K_E^\perp \cong A_1 \oplus A_2$, the same rotation is -1 on A_1 and a Coxeter element, with characteristic polynomial $\lambda^2+\lambda+1$, on A_2. Equivariance now identifies the line with A_1 and the plane with A_2. These finite divisor and edge calculations are also reproduced by `branch_locus.py`. $\square$

Theorem 8.8. *Let X be in the admissible framework, with $\omega_X \cong \mathcal{O}_X$, $h^1(\mathcal{O}_X) = 0$, and the $\partial\bar\partial$-lemma for a resolution. Suppose there is a class*

$$\kappa \in \ker\Big(\bigoplus_p H_2(L_p;\mathbb{Q}) \longrightarrow H_2(\widehat{X};\mathbb{Q})\Big), \qquad \kappa_p \in \mathcal{R}_p \smallsetminus \{0\} \ \textit{ for every } p.$$

Then X admits a first order smoothing $\theta \in \mathbb{T}^1_X$.

The hypothesis is exactly the criterion of [Wue26b]: the kernel is its relation space, the condition $\kappa_p \in \mathcal{R}_p$ is its branch restriction, and $\kappa_p \neq 0$ is its nonvanishing. So Theorem 8.8 is the converse of [Wue26b, Thm. 6.1] at first order, and by [Wue26b, Rem. 6.2] its hypothesis is equivalent to the criterion being silent.

Proof. Let $E = \mathrm{Exc}(\widehat{X} \to X)$ and $V = \widehat{X} \smallsetminus E = X_{\mathrm{reg}}$. Excision and the local cohomology sequence give

$$H^1(V;\Omega^2_V) \longrightarrow \bigoplus_p H^2_{E_p}(\widehat{X}_p;\Omega^2_{\widehat{X}_p}) \longrightarrow H^2(\widehat{X};\Omega^2_{\widehat{X}}). \tag{11}$$

By Lemma 8.6, κ determines a class $\widetilde{\kappa} \in \bigoplus_p K_p^{\mathrm{cr}}$ in the middle term.

The compatibility assertion in that lemma gives the commutative square

$$\begin{array}{ccc} \bigoplus_p H_2(L_p;\mathbb{C}) & \longrightarrow & H_2(\widehat{X};\mathbb{C}) \\ \downarrow & & \downarrow \mathrm{PD} \\ \bigoplus_p K_p^{\mathrm{cr}} & \longrightarrow & H^4(\widehat{X};\mathbb{C}), \end{array}$$

where the lower arrow is the map in (11) followed by the de Rham map, and the upper arrow is induced by the inclusions of the links. The hypothesis on κ therefore says that the de Rham image of $\widetilde{\kappa}$ is zero. The $\partial\bar\partial$-lemma makes $H^2(\widehat{X};\Omega^2_{\widehat{X}}) \to H^4(\widehat{X};\mathbb{C})$ injective, so $\widetilde{\kappa}$ already maps to zero in the last term of (11). Exactness supplies $\eta \in H^1(V;\Omega^2_V)$ mapping to $\widetilde{\kappa}$.

A global trivialisation of ω_X identifies $H^1(V;\Omega^2_V)$ with $H^1(V;\Theta_V)$. Schlessinger's global comparison, in the depth-3 form of [FL25, Rem. 4.2], identifies the latter with $\mathbb{T}^1_X$. Let θ be the class corresponding to η. Restricting (11) to a neighborhood of p shows that the local component of θ and the class corresponding to κ_p have the same image in K_p^{cr}; the local connecting map is injective by Lemma 8.6, so they are equal. Lemma 8.7 and the assumptions $\kappa_p \in \mathcal{R}_p \smallsetminus \{0\}$ now put every local component of θ in $\mathrm{Sm}_p \smallsetminus \{0\}$. Thus θ is a first order smoothing. $\square$

Remark 8.9. What Theorem 8.8 does not give is integrability. Locally there is no obstruction, since the smoothing components are smooth; what is not automatic is that θ lifts to a one-parameter deformation of X. Theorem 8.2 settles the $\mathbb{Q}$-factorial case, but no X in the admissible framework carrying one of the prescribed singular faces is $\mathbb{Q}$-factorial, so that settles nothing for the singular case and the question is the singular part of the framework. Nor is the usual counterexample decisive: [Gro97r, Ex. 2.19] is built from a germ that does not smooth at all, and so lies outside the framework. What is missing is an unobstructedness theorem for isolated canonical Gorenstein singularities that are not complete intersections, and Conjecture 8.1 reduces to exactly that.

Remark 8.10. Applied to Δ_{19}, whose criterion is silent, Theorem 8.8 says X_{19} *has* a first order smoothing. Theorem 5.2 says no single-degree family of the ambient produces one, at any degree. Both hold: the deformation exists infinitesimally and is invisible to the toric mechanism.

8.5. **What remains, precisely.** Write $\mathbb{T}^1_X = \operatorname{Ext}^1(\Omega^1_X, \mathcal{O}_X)$ and let

$$0 \to H^1(\Theta_X) \to \mathbb{T}^1_X \xrightarrow{\alpha} H^0(\mathcal{T}^1_X) = \bigoplus_i T^1_{X,x_i} \xrightarrow{\text{ob}} H^2(\Theta_X)$$

be the low-degree exact sequence of the local-to-global Ext spectral sequence, and let $\Phi\colon \bigoplus_i T^1_{X,x_i} \to \operatorname{Cl}(\mathbb{P}_\Delta)^*_{\mathbb{Q}}$ be the map of [Wue26b, Prop. 8.2], whose kernel is the relation space. Exactness gives $\operatorname{im}\alpha = \ker \text{ob}$. The two maps ob and Φ have different targets, so the precise comparison is an equality of kernels, not a literal equality of maps. Theorem 8.8 proves $\ker\Phi \subseteq \ker \text{ob}$ through the support sequence, while naturality of the same sequence sends every global first-order class to zero under the topological map, giving the reverse inclusion. Thus a section of $\mathcal{T}^1_X$ lifts to a first-order deformation of X if and only if the total curve class it determines vanishes. What then remains is integrability, and that is not a formality. Gross obtains unobstructedness only when $T^2_{\text{loc}} = 0$, which for an isolated singularity means complete intersection, and a cone over a del Pezzo surface of degree $d > 4$ is never one [Gro97, (3.3)].

The example usually cited against unobstructedness here does not in fact reach our germs, and saying so sharpens the question rather than settling it. Gross's non-reduced $\operatorname{Def}(X)$ [Gro97r, Ex. 2.19] is built from cones over $\mathbb{F}_1$, of degree 8; by [Gro97, (5.5)] that germ has $\dim T^1 = 1$ and *no smoothing at all*, its versal base being a fat point, and the non-reducedness of the global deformation space is inherited from it. Our germs are the degree-6 and degree-7 rows of the same list, where the versal base is positive-dimensional and every smoothing component is smooth. So the standard counterexample is outside the framework, for the structural reason that it does not smooth; what is missing is a positive theorem, not a way past a negative one.

Remark 8.11 (X_{19} is the sharp test). Conjecture 8.1 predicts that X_{19} *is* smoothable, since the criterion of [Wue26b] is silent on it. Theorem 5.2 says that the one uniformly available construction does not produce that smoothing, at any degree. So X_{19} is where the conjecture is most exposed: a proof of Conjecture 8.1 must produce a smoothing of X_{19} by a mechanism Theorem 5.2 has already excluded, and a demonstration that X_{19} is non-smoothable would refute the conjecture outright. Theorem 5.2 is thus better read as a localisation of the difficulty than as a negative result.

Appendix A. Two routes with no input here

Both of the following look like alternatives to the mechanism of Section 3, and neither has anything to act on in the admissible framework. We record them because the second in particular is the natural first thing to try.

A.1. **Mavlyutov's non-polynomial deformations.** Mavlyutov [Mav04, Mav05] showed that a Batyrev hypersurface can have deformations not induced by the ambient linear system, and counted them: the non-polynomial deformations are counted by

$$\sum_{\operatorname{codim}\Gamma=2} \ell^*(\Gamma)\,\ell^*(\Gamma^*), \tag{12}$$

where ℓ^* is the number of relative-interior lattice points and Γ^* runs over the dual faces, which for a codimension-two face Γ of Δ° is an *edge* of Δ.

Proposition A.1. *If every edge of Δ has lattice length one, in particular if Δ is in the admissible framework, then* (12) *vanishes.*

Proof. $\ell^*(\Gamma^*) = 0$ for every edge Γ^* of lattice length one, so every term vanishes. □

The isolated-singularity hypothesis and the vanishing of (12) are therefore the same condition read twice. The mechanism is structurally empty exactly where our germs are isolated.

A.2. **Global Minkowski decomposition.** The other construction in [Mav05, §7] takes as input a Minkowski decomposition of Δ itself.

Proposition A.2. *Δ_{19} is Minkowski-indecomposable: its only weak Minkowski summands are its own homothets.*

Proof. The weak Minkowski summands of a polytope are the solutions of the edge-cycle system (6) imposed simultaneously on all of its 2-faces, one vector equation per 2-face in the edge dilations. Δ_{19} has f-vector $(19, 64, 72, 27)$, so the system has 64 unknowns and 72 vector equations, and its solution space is one-dimensional, spanned by the all-ones vector, which is the homothety direction. A polytope whose summand space is a single ray is indecomposable. The rank computation is exact over $\mathbb{Q}$ in `global_decomp.py`, which also verifies the f-vector against Euler's relation. □

So that route has no nontrivial input on Δ_{19}, independently of the status of the unproved statements in [Mav05, §7].

A.3. **Why one cannot simply refine to a simplicial ambient.** It is natural to try to avoid the non-simplicial ambient by passing to a maximal projective crepant partial (MPCP) desingularisation, which cones off every lattice point of Δ. This destroys the datum.

Over a singular 2-face G, an MPCP subdivision refines σ_G using the interior lattice points of G; at a vertex degree the slice cell *is* G, so the cell is cut into triangles. A triangle is Minkowski-indecomposable, so on the refined complex every cell over G has only homothetic summands, and Definition 3.1(D1) makes the induced decomposition trivial precisely where the germ is. On Δ_9 the two node squares have no interior point and survive untouched, while the pentagon has one and its cell becomes five triangles: *the segment-plus-triangle datum is destroyed, and only that one.*

Structurally this is not an accident. The crepant subdivision *resolves* the divisorial germ, and Altmann's deformation is the alternative to resolving it. One cannot have both, and any argument of the kind in Section 7 must work on the coarse, non-simplicial ambient.

Appendix B. On the literature this rests on

Two steps above use published statements whose proofs we could not locate in the form used. Each is recorded here rather than left for a citation to carry.

First, [Sü08, Prop. 3.1] asserts in the first sentence of its proof that a polyhedral divisor with complete locus and two coefficients *describes the affine toric variety X_δ*, citing [AH06, §11]. Section 11 of that paper contains the *downgrade*, that is, given a subtorus of the big torus of a toric variety, produce the polyhedral divisor, and not the converse in this form. We use the assertion in Proposition 7.11.

Second, no source we consulted states that the cones obtained by upgrading the charts of a complexity-one T-variety assemble into a fan. Süß [Sü08, Rem. 1.6] says only that each polyhedral divisor "gives rise to a cone"; [IV11, Rem. 2.4] asserts that the upgraded p-divisors form a divisorial fan, but it is an unproved remark, carries hypotheses (contraction-freeness, smoothness of the base), and is never specialised to the case of a point base, which is what would turn a divisorial fan into a fan of cones. This is exactly why Proposition 7.11 is proved chart by chart rather than by assembling a global fan for the general fibre: the assembly step is not available.

We also note three discrepancies between the sources that a reader reconstructing these computations will meet. The ambient ordering is transposed between [Sü08] (base coordinate first) and [IV11] (fibre first); whether to take a closure and whether to adjoin the tail cone differ between [Sü08, Prop. 3.1], [Sü08, Rem. 1.6] and [IV11, Prop. 2.1]; and the downgrade formula in [Sü08, Rem. 1.6] as printed omits the intersection with the cone that [AH06, §11], [AHS08, §5] and [AIPSV11, §4.2] all include. None of these affects the results above, all of which were computed from the versions carrying the intersection.

Appendix C. Certificates

Every numerical claim above is produced by a script in the accompanying repository. Each computes its input from the polytope's vertex list and prints the individual checks it passes. We list them against the statements they certify.

statement	script	checks
Rem. 4.3, 4.4	`one_facet.py`	7
Prop. 4.10, Def. 4.12, Rem. 4.8	`locking.py`	13
Thm. 5.2, the line $\mathcal{A}_P$, its two ranges, and cell rigidity	`slice_rigidity.py`	49
independent reimplementation	`slice_rigidity.sage`	24
face compatibility in Corollary 4.14	`def41_check.sage`	16
Prop. A.2	`global_decomp.py`	12
Section 6, the sweep	`dp7_facet_sweep.py`	n/a
what the survivors induce	`candidates.sage`	n/a
Appendix A.3	`refine.py`	5
Props. 7.1, 7.2	`v09_candidate.py`, `sing_locus.py`	6
Prop. 7.3, Cor. 7.4	`slice_admissible.sage`	36
Cor. 7.4, completeness of the general fibre	`general_fibre.sage`	6
Prop. 7.7; independent semiampleness diagnostic	`hzero.sage`	n/a
Prop. 7.8	`toric_fibre.sage`	5
Prop. 7.11	`charts.sage`	3
Rem. 7.14	`final_check.sage`	3
Sm_p corresponds to $\mathcal{R}_p$, both germ types	`branch_locus.py`	17
and why the cyclic criterion fails here	`smoothing_locus.py`	13
Thm. 6.3, and [Wue26b, Prop. 9.2]	`classification.py`†	10
the relation classes of X_9 and X_{19}	`relation_class.py`†	36
from Theorem 7.12's family	`theoremB_class.py`	10

The older scripts `gate1_final.sage` and `gate1_caseC.sage` are deliberately not listed as certificates. They test smoothness of cones over candidate summands but do not impose admissibility (D2), so they cannot certify a necessary list of degrees. The diagnostic `gate1_admissible.sage` exhibits this failure on the candidate pairing vectors. The proof now uses Lemma 4.6 instead; its only finite calculation is the exact line and the two pairing formulas certified by `slice_rigidity.py`.

The two marked † certify statements of [Wue26b] as well as statements here, and live in that paper's certificate directory of the shared repository, from which the scripts of this paper already import their exact toric modules. Two further scripts there, `kernel_equality.py` and `lone_germ.py`, certify [Wue26b, Prop. 8.3] and [Wue26b, §9.2], both quoted above.

`slice_rigidity.py` and `slice_rigidity.sage` implement the same statement in different systems, pure Python with an integer forcing chain and Sage with convex hulls over $\mathbb{Q}$ and then over $\mathbb{Q}(s)$, and are compared against each other and against an independent facet computation.

`hzero.sage` is validated on the worked example of its source before use, reproducing the printed value $h^0(-K_{\mathbb{P}(\Omega_{\mathbb{P}^2})}) = 27$, and then cross-checked on the special fibre against the toric lattice-point count.

`charts.sage` carries a control: run on the special fibre, where the answer is known, it must return positive-dimensional singular loci in exactly the four charts containing a singular 2-face. It does.

`theoremB_class.py` carries a test the construction could have failed. Converting Theorem 7.12's Minkowski data germ by germ into elements of T^1 by the jump formula of [Wue26b, Prop. 8.2], the three directions turn out to be linearly dependent with every coefficient nonzero, uniquely up to scale, and nothing forces that, since the closing conditions are imposed one 2-face at a time. So the family produces the relation class that [Wue26b] predicts, computed from the deformation rather than read off the matrix. Its dP_7 component lies on the (-2)-line with zero component on the complement, which is the branch restriction [Wue26b] imposes as a hypothesis.

`locking.py` carries the control that motivates Remark 4.8: it searches the facets of Δ_{19} for one that is not locked and whose Minkowski summand cone changes dimension under rescaling, and finds facet 12, with dimension 2 at the lattice facet and 1 elsewhere. It also checks that a rule firing on opposite rather than consecutive edges would call the cube rigid. The two implementations of the propagation, in `locking.py` and in `dp7_facet_sweep.py`, are compared facet by facet and agree on all 27 facets of Δ_{19}.

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